Arithmetic Expression Geometry III: Singular Zero Geometry and Tubes

Branched Pullbacks, Arithmetic Zero Networks, and Topological Transport

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Abstract

Arithmetic expression geometry associates a real assignment to an oriented Riemannian surface and imposes the eikonal identity $|\nabla a|_g^2 = \mu^2 + \lambda^2 a^2$, with $\mu \neq 0$. Its regular zeros are curves, not isolated roots. We develop their singular and parameterized zero geometry while keeping real zero tubes distinct from complex finite-root braids.

Conformal rescaling makes every surface submersion a regular AES. Holomorphic pullback is more rigid: local degree m produces a $2m$-pronged zero and cone angle $2\pi m$. Applied to the normalized Hauptmodul of the $(2, 4, \infty)$ Hecke group generated by $z \mapsto z + \sqrt{2}$ and $z \mapsto -1/z$, it gives a singular AES whose zero dessin, after bivalent suppression, is the $\{\infty, 4\}$ network with four-valent 4π cone nodes.

Its sign character selects a cover of signature $(0; 2; \infty, \infty)$ whose coarse completion is the complete AEG cylinder $\mathbb{C}^\times$. Retaining the hyperbolic orbifold structure and using the standard cusp compactification gives $T^1\mathcal{O}_4 \cong \mathbb{RP}^3 \setminus K_\infty$ and $T^1\mathcal{O}_4^\chi \cong S^3 \setminus T(2, 4)$. Primitive hyperbolic classes give periodic-orbit knots; the zero dessin is their coding spine, and a reduced code satisfies $\mathrm{Lk}(K_\gamma, K_\infty) = \sum_k (j_k - 2)$. The geodesic-flow and linking theorems are classical; the synthesis identifies the AEG sign cover with their natural two-fold topological cover, not a new knot invariant.

We test history-to-divisor naturality on typed registers. The relation $u^2 = t$ forces an equivariant quadratic endpoint divisor for every projective history. Its abstract cover and Frobenius cycles are operator-independent, and its terminal correspondence factors through projective evaluation; a tagged prefix trace and pole divisor nevertheless distinguish Paper I's neutral word ω_4 from the empty history. The registers $v^2 = 3$, $u^2 = t + v$ force $(u^2 - t)^2 - 3$, arithmetically irreducible but split into two constant-geometric components, with explicit discriminant, monodromy, and Frobenius. This gives a minimal decorated model and, within it, a terminal-divisor no-go; the unrestricted functor remains open.

A parameterized zero-section theorem separates transversality from properness. Compact boundaryless oriented real zero tubes over a circle are unions of tori; explicit helical and logarithmic families compute their transport. Singular Morse models realize topology change, whereas square-free complex root families realize every braid. Finally, affine conjugation recovers an Alexander quandle: its ordinary torsion is exact, and a nonzero resonant twisted class has planar state sum equal only to the coloring count. Functorial thread selection, Markov descent, and separation beyond Alexander or Burau remain open.

Keywords: arithmetic expression geometry; branched pullback; Hecke triangle group; Hauptmodul; geodesic knot; torus link; relative divisor; Frobenius; singularity; braid group

Contents

1 Singular AEG and the hierarchy of zero objects

1.1 The question

Paper I associates to continuous affine evaluation a regular arithmetic expression space

$$(M, g, a; \mu, \lambda), \qquad |\nabla a|_g^2 = \mu^2 + \lambda^2 a^2, \qquad \mu \neq 0, \tag{AEG}$$

where M is an oriented smooth surface, g is Riemannian, and a is a real-valued assignment function [18]. The right-hand side of (AEG) is strictly positive. Thus a is a submersion, and every nonempty level set—in particular $a^{-1}(0)$—is a regular curve. An isolated zero, a crossing, a branch point, or a birth/death event cannot be a regular interior zero.

The word *tube* is nevertheless natural. If an assignment varies with a parameter, its zero curves sweep a surface in the product of the arithmetic surface and the parameter space. The central problem is to determine when that swept surface is more than a picture: when it is smooth, when its projection is a fiber bundle, where topology may change, and what additional data are needed before a braid or knot can be extracted.

There is a second problem, logically prior to the tube: why should a complicated zero set be attached to arithmetic expressions at all? Conformal realization can manufacture almost any regular zero pattern, so geometric abundance alone does not answer this question. The main analytic–arithmetic construction of this paper starts instead from the projective operators of Paper I. Translation by $\sqrt{2}$ and negative inversion generate the $q = 4$ Hecke group; its Hauptmodul then produces a branched zero network whose four-valent vertices are forced by the elliptic ramification. This supplies an exact model in which arithmetic operator paths, holomorphic branching, and singular AEG geometry meet.

The historical AEG notes used "tube" for several disjoint unions of ambient geometries. A source audit of the repository and its Git history found no verifiable general construction under the old symbols E_k or $E_{\log}$. Accordingly, this paper does not import those symbols as established objects. It constructs descriptive finite-zero, logarithmic-cover, and helical models from first principles. The distinction matters: an ambient family is not its zero incidence, and a smooth zero incidence is not automatically a tube.

1.2 Imported interfaces and new scope

We use four results of Papers I and II as interfaces rather than redefining them.

1. Paper I proves the regular-zero theorem just recalled and defines a *singular AES*. Its singular set is closed, nowhere dense, and locally essential, and the regular equation holds on the complement.

2. Paper I proves that spatial regularity along the total zero set of a smooth real family makes that total zero set a smooth submanifold and its parameter projection a submersion. It explicitly leaves global triviality open without properness.

3. Paper II equips every oriented regular AES with its underlying complex structure. On the complete basic hyperbolic model, the arithmetic-holomorphic coordinate $w = X + iy$ is a global complex coordinate, and arithmetic holomorphicity is ordinary holomorphicity in w [19].

4. Paper I's bilateral projective evaluation contains translations and negative inversion in $\mathrm{PGL}_2(K)$. Over $K = \mathbb{Q}(\sqrt{2})$ these operators generate the Hecke subgroup used in Section 4. This import is at the operator quotient level and does not identify different raw history words or ordinary arithmetic domains [18].

The first two interfaces concern a real scalar assignment and hence codimension-one zero curves. The third permits complex-valued auxiliary fields and hence codimension-two isolated roots. They must not be conflated.

Definition 1.1 (Stratified singular AES)**.** A *stratified singular AES* is a singular AES in the sense of Paper I together with a specified locally finite Whitney stratification $\mathcal{S}$ of its singular set S. A stratum is called a *singular zero stratum* if it is contained in the closure of $a^{-1}(0) \cap (M \setminus S)$. The stratification is additional control data; it is not claimed to be canonical.

This refinement supplies a language for parameterized Morse examples and branched fillings. It does not solve the classification of singular AEG germs. Indeed, the conformal realization theorem in Section 2 shows that the bare singular-AES category is deliberately flexible.

1.3 Seven levels that must remain distinct

Let $q : X \to B$ be an ambient family and let s be a real or vector-valued field on X. The following hierarchy fixes our terminology.

Definition 1.2 (Zero-object hierarchy)**.** Write $\mathcal{Z}(s) = s^{-1}(0)$ and $\pi = q|_{\mathcal{Z}(s)}$.

1. $\mathcal{Z}(s)$ is the *total zero set*; no regularity is implied.
2. It is a *smooth zero incidence* if it is a smooth submanifold of X.
3. It is a *proper zero tube* if π is a surjective, proper smooth submersion (with the corresponding neat boundary hypotheses when boundaries are present).
4. It is an *embedded zero tube* when its inclusion in the specified ambient family is retained as part of the data.
5. A *threaded zero tube* is an embedded zero tube together with a specified finite multisection or other explicitly declared one-dimensional thread datum.
6. A *braid closure* is obtained only after a finite moving-point system, an ambient surface, and a closure convention have been fixed.
7. A *knot invariant* is a quantity on closures that has been proved independent of all presentation choices, in particular the appropriate conjugation and stabilization moves.

The hierarchy also exposes a dimension obstruction. A real field on a surface has one-dimensional zero fibers; a complex field has zero-dimensional root fibers. Only the latter produces an ordinary geometric braid without an extra transverse probe.

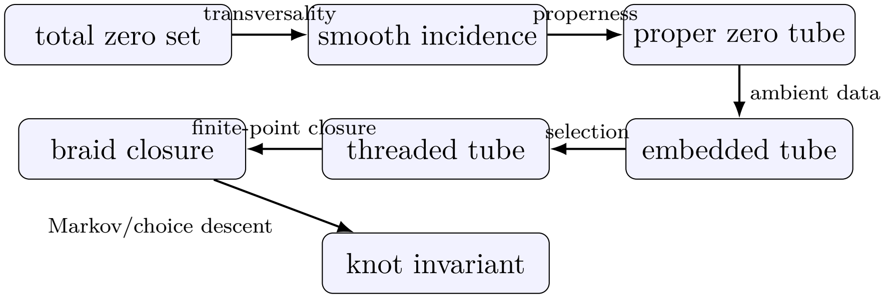

Figure 1: Each arrow requires new hypotheses or data. No arrow in the diagram is automatic from the object to its left.

1.4 Principal results

The paper proves the following claims. Status qualifications are included to make the mathematical boundary visible.

The positive and negative results are complementary. The realization theorems show that tube and braid geometry genuinely occur inside AEG. The flexibility and collapse theorems show that occurrence alone carries little discriminatory power. Any new invariant must come from a natural decoration tied to expression history, and that decoration must survive the relevant quotient.

1.5 Organization

Section 2 proves conformal and holomorphic pullback realization, including finite Blaschke and essential-singularity controls. Section 3 records the flexible parallel and logarithmic models. Section 4 is the arithmetic core: it constructs the $q = 4$ Hecke–Hauptmodul singular AES, proves its character-valued descent, and separates zero histories from relative prime divisors. Section 5 identifies the sign cover, its cusp link, and the precise coding-spine interface to Hecke periodic-orbit knots, without identifying the two-dimensional zero graph with a three-dimensional template. Section 6 tests divisor naturality on explicit finite register covers and records the exact information lost at the operator quotient. Sections 7–9 develop parameterized incidence, proper tubes, and singular change. Section 10 treats rank-two root monodromy, and Section 11 applies the Markov and Alexander/Burau novelty gates. The logical order is therefore

$$\begin{aligned} &\text{branched analytic germ} \longrightarrow \text{arithmetic zero network} \\ &\longrightarrow \text{relative divisor and discriminant} \\ &\longrightarrow \text{tube and monodromy.} \end{aligned}$$

1.6 Conventions

All manifolds are smooth, Hausdorff, and second countable. Unless boundary hypotheses are stated, manifolds are boundaryless. The parameters $\mu \neq 0$ and $\lambda \in \mathbb{R}$ always belong to the AEG equation; family parameters are denoted by b, t, θ, or τ. The letter t is also used for an affine multiplier only in Section 11, where it cannot be confused with a family coordinate. A deck transformation is denoted D, while T_{coef} denotes the Laurent-module action in twisted quandle cohomology.

For a loop of configurations, a based loop determines a braid element and a free loop determines a conjugacy class. "Invariant" without qualification means an ambient-isotopy invariant of oriented links; annular, transverse, framed, and presentation-dependent data are named explicitly.

2 Local zero models and conformal realization

2.1 The regular germ

At a regular zero p, the submersion theorem gives coordinates (u, v) centered at p in which $a = \mu u$: take $u = a/\mu$ and choose the sign of v so that (u, v) is positively oriented. This statement concerns the zero germ and the assignment; it does not put the metric into Euclidean form. Equation (AEG) further gives $|\nabla a|_g = |\mu|$ along $a = 0$. Thus a regular zero has one transverse direction with a fixed nonzero speed and one tangent direction. No finite-order singularity can occur there.

The simplest global realization keeps the metric fixed.

Example 2.1 (Cylindrical propagation)**.** On $M = \mathbb{R}_r \times S^1_\phi$ with $g = \mathrm{d}r^2 + \mathrm{d}\phi^2$, set

$$a_c(r, \phi) = \begin{cases} \dfrac{\mu}{\lambda} \sinh(\lambda(r - c)), & \lambda \neq 0, \\ \mu(r - c), & \lambda = 0. \end{cases}$$

Then $|\nabla a_c|_g^2 = \mu^2 + \lambda^2 a_c^2$ and $Z(a_c) = \{c\} \times S^1$. This model will yield a proper product tube in Section 8.

2.2 Conformal realization

The following construction is elementary, but it determines the scope of every subsequent singularity claim.

Theorem 2.2 (Conformal realization)**.** *Let (M, h) be an oriented Riemannian surface, let $a \in C^\infty(M, \mathbb{R})$ be a submersion, and fix $\mu \neq 0$ and $\lambda \in \mathbb{R}$. The conformal metric*

$$g_a = \frac{|\,\mathrm{d}a|_h^2}{\mu^2 + \lambda^2 a^2}\, h \tag{1}$$

is smooth and positive definite, and $(M, g_a, a; \mu, \lambda)$ is a regular AES.

Proof. Put $g_a = \rho h$. Since $\mathrm{d}a$ never vanishes and the denominator is positive, ρ is a positive smooth function. The inverse metric is $g_a^{-1} = \rho^{-1} h^{-1}$, so

$$|\nabla a|_{g_a}^2 = |\,\mathrm{d}a|_{g_a^{-1}}^2 = \rho^{-1} |\,\mathrm{d}a|_h^2 = \mu^2 + \lambda^2 a^2.$$

□

Corollary 2.3 (Singular realization)**.** *Let $a \in C^\infty(M, \mathbb{R})$ have a closed nowhere-dense critical set $S = \mathrm{Crit}(a)$. On $M \setminus S$, define g_a by (1). Then $(M, S, g_a, a; \mu, \lambda)$ is a singular AES. Every point of S is locally essential.*

Proof. Theorem 2.2 applies on $M \setminus S$. If a point $p \in S$ admitted a regular local extension agreeing with a, then $\mathrm{d}a_p = 0$ would imply $|\nabla a|^2(p) = 0$ for the extended nondegenerate metric, whereas the right-hand side of (AEG) is at least $\mu^2 > 0$. This is impossible. □

2.3 Holomorphic branched pullback

Paper II identifies arithmetic holomorphicity with ordinary holomorphicity in the complex structure carried by an oriented regular AES. The next theorem uses that interface, but the singular metric and its local completion are proved here. The statement therefore does not depend on an unproved singular extension of the function theory.

Theorem 2.4 (Branched-pullback zero germ)**.** *Let Σ be a Riemann surface, let $\Phi : \Sigma \to \mathbb{C}$ be a nonconstant holomorphic map, and write $\Phi = U + iV$. Fix $\mu \neq 0$ and $\lambda \in \mathbb{R}$. Put*

$$a_\Phi = V, \qquad g_\Phi = \frac{|\,\mathrm{d}\Phi|^2}{\mu^2 + \lambda^2 V^2} \quad \text{on } \Sigma \setminus \mathrm{Crit}(\Phi), \tag{2}$$

where $|\,\mathrm{d}\Phi|^2 = |\Phi'(z)|^2|\,\mathrm{d}z|^2$ in a holomorphic coordinate. Then

$$(\Sigma, \mathrm{Crit}(\Phi), g_\Phi, a_\Phi; \mu, \lambda)$$

is a singular AES. Every critical point is locally essential.

Suppose $p \in \mathrm{Crit}(\Phi)$, $\Phi(p) \in \mathbb{R}$, and the local degree of Φ at p is $m \geq 2$. Then:

1. *the zero germ $a_\Phi^{-1}(0)$ has exactly $2m$ half-branches, equally spaced in a suitable holomorphic coordinate;*
2. *the metric completion of the punctured germ adds one cone point of angle $2\pi m$;*
3. *p is a critical zero of the smooth assignment a_Φ, and no smooth nondegenerate regular-AES metric can extend across it.*

For $m = 2$ this is a four-pronged zero with cone angle 4π.

Proof. On the target plane consider

$$b(U + iV) = V, \qquad h_{\mathrm{lin}} = \frac{\mathrm{d}U^2 + \mathrm{d}V^2}{\mu^2 + \lambda^2 V^2}.$$

It satisfies $|\nabla b|^2_{h_{\mathrm{lin}}} = \mu^2 + \lambda^2 b^2$. Away from $\mathrm{Crit}(\Phi)$, formula (2) is exactly $\Phi^* h_{\mathrm{lin}}$, and $a_\Phi = b \circ \Phi$. A holomorphic local diffeomorphism preserves the eikonal identity under pullback, proving regularity on the complement.

At a critical point, $\mathrm{d}a_\Phi = 0$. A smooth nondegenerate extension satisfying the regular AEG equation would give $0 = |\nabla a_\Phi|^2 \geq \mu^2$, which is impossible. Thus every critical point is locally essential in the sense of Paper I.

At a critical zero of local degree m, the holomorphic local-mapping theorem provides coordinates centered at p and $\Phi(p)$ in which

$$\Phi(z) - \Phi(p) = z^m.$$

Consequently the zero germ is

$$\mathrm{Im}(z^m) = 0,$$

which consists of the $2m$ rays $\arg z = k\pi/m$, $k = 0, \ldots, 2m-1$. The target metric is smooth and positive at $\Phi(p)$. If it is written locally as $e^{2\psi(w)}|\,\mathrm{d}w|^2$, its pullback is

$$m^2|z|^{2m-2}e^{2\psi(z^m)}|\,\mathrm{d}z|^2.$$

For a circle $|z| = r$, the ratio of its length to the radial distance from the missing point tends to $2\pi m$. The completion therefore has cone angle $2\pi m$. The last assertion is the essentiality argument already given. $\square$

Remark 2.5 (Pullback naturality and its limit)*.* The theorem is more structured than arbitrary conformal realization: the metric, assignment, zero germ, and cone angle are all pulled back from one holomorphic map. It still does not make every such map arithmetic-natural. A history-derived source for the map is an additional requirement, supplied at the projective operator level by the Hecke construction in Section 4 and left open in greater generality.

Example 2.6 (Finite Blaschke bridge)**.** Let

$$B(z) = e^{i\theta} \prod_{j=1}^{n} \frac{z - \alpha_j}{1 - \overline{\alpha_j} z}, \qquad |\alpha_j| < 1,$$

be a finite Blaschke product. The single holomorphic map $B : \mathbb{D} \to \mathbb{D}$ supports three related zero objects.

First, $B^{-1}(0)$ is an effective finite divisor of degree n, counted with multiplicity. A family of such products whose roots remain distinct produces the finite root incidence of Section 10. Second, for every phase ϕ, the real assignment

$$a_\phi = \operatorname{Im}(e^{-i\phi} B), \qquad g_\phi = \frac{|B'(z)|^2}{\mu^2 + \lambda^2 a_\phi^2} |\,\mathrm{d}z|^2$$

has zero network

$$Z(a_\phi) = B^{-1}(e^{i\phi}\mathbb{R}).$$

At a critical point of local degree m lying over this diameter, the network has $2m$ prongs and the metric completion has cone angle $2\pi m$.

Third, pull back the isolated-zero disc of Paper I, Equation (4), along B. The assignment $a_{\mathbb{D}} \circ B$ has the finite isolated zero set $B^{-1}(0)$, while $B^* g_{\mathbb{D}}$ has cone angle $2\pi m$ at a critical point of local degree m. The declared singular set is $B^{-1}(0) \cup \operatorname{Crit}(B)$. At a simple root the pulled-back radial assignment is nonsmooth. At a multiple root an even power of the radius can become smooth, but the zero still cannot be a regular AES point: its differential vanishes and the pulled-back metric is degenerate there. Thus one holomorphic object links an isolated divisor, phase-dependent real zero networks, and, after parameterization, finite threads or braids.

The roots α_j are arbitrary data. This bridge is a finite universality mechanism, not a proof that an expression history canonically selects B.

Example 2.7 (An analytic-capacity control)**.** Let $M = \{|z| < \varepsilon\}$ and, for $z \neq 0$, set

$$\Phi(z) = \sin(1/z), \qquad a(z) = \operatorname{Im}\Phi(z), \qquad g = \frac{|\Phi'(z)|^2}{\mu^2 + \lambda^2 a(z)^2} |\,\mathrm{d}z|^2.$$

Assign any value, say $a(0) = 0$, at the origin and declare

$$S = \{0\} \cup \left\{ z_k = \frac{1}{\pi/2 + k\pi} : k \in \mathbb{Z},\ |z_k| < \varepsilon \right\}.$$

Since $\Phi'(z) = -\cos(1/z)/z^2$, this is precisely the singular set in M. Writing $z = x + iy$ and $c_k = \pi/2 + k\pi$, the zero set away from the origin is

$$\begin{aligned} Z(a) = {} & \{y = 0\} \\ & \cup \bigcup_{k \in \mathbb{Z}} \left\{ \left(x - \frac{1}{2c_k}\right)^2 + y^2 = \frac{1}{4c_k^2} \right\}, \end{aligned} \tag{3}$$

with all sets restricted to $0 < |z| < \varepsilon$. Indeed, $\operatorname{Im}\sin(u+iv) = \cos u \sinh v$, and inversion sends the vertical lines $u = c_k$ to the displayed circles. Each circle meets the real axis at the simple critical point z_k, where Theorem 2.4 gives a four-pronged zero and cone angle 4π. The nodes and circles accumulate at the essential singularity 0. Indeed, $\operatorname{Im}\sin(1/z)$ is unbounded in every punctured neighborhood of the origin, so the arbitrarily assigned value $a(0)$ admits no continuous, hence no regular-AES, local extension.

This example proves analytic capacity: singular AEGs can carry an infinite four-valent zero network. It is not promoted to the arithmetic model of the paper, because the formula does not distinguish expression histories or induce a canonical arithmetic quotient.

Warning 2.8 (Flexibility boundary). When the metric may be chosen as part of the construction, the regular AEG equation admits every smooth submersion, and the singular category admits every smooth function with closed nowhere-dense critical set. Therefore the bare equation does not by itself select an AEG-specific list of singularity types. A classification theorem would require additional naturality or analytic hypotheses, such as a fixed ambient metric, completeness, curvature bounds, controlled metric asymptotics, homogeneity, or derivation from expression history.

The warning is not a defect of the construction. It is a precise division of labor. The equation controls assignment speed relative to a chosen metric; it does not select that metric. Later tube theorems are consequently topological theorems with explicit hypotheses, not rigidity theorems for arbitrary AEGs.

2.4 Singular zero mechanisms

There are at least four mechanisms by which the regular-zero conclusion can fail.

1. The assignment may fail to be differentiable at a point, even though the metric extends there.
2. The assignment may be smooth but critical, forcing the conformal realization metric to degenerate or be removed at that point.
3. The domain may have a boundary or a puncture through which a zero component enters, exits, or escapes.
4. In a parameterized family, the total incidence may remain smooth while its projection to parameter space ceases to be a submersion.

Paper I supplies a verified example of the first mechanism. On the unit disc with $\rho = (\xi^2+\eta^2)^{1/2}$, it takes

$$g_{\mathbb{D}} = \frac{4}{\lambda^2(1-\rho^2)^2}(\,\mathrm{d}\xi^2 + \mathrm{d}\eta^2), \qquad a_{\mathbb{D}} = \frac{2\mu}{\lambda}\frac{\rho}{1-\rho^2} \quad (\rho > 0), \tag{4}$$

with $a_{\mathbb{D}}(0) = 0$ and $\mu, \lambda > 0$. The metric extends smoothly, the assignment is not C^1 at the center, and the center is its only zero. This is an isolated singular zero, not a normal form for all isolated singular zeros.

The second and fourth mechanisms will appear together in the Morse families of Section 9. The third is the reason properness and boundary control occur explicitly in Definition 1.2.

2.5 What local equivalence should preserve

For clarity, we distinguish three notions that are sometimes compressed into the phrase "same singularity."

Definition 2.9 (Three local equivalences)**.** For two pointed singular AEGs, a *zero-germ equivalence* is a local diffeomorphism carrying one zero germ to the other. An *assignment-germ equivalence* additionally intertwines the assignments up to a local diffeomorphism of the target fixing zero. An *AEG-germ isomorphism* preserves the assignment, the oriented metric, and the constants (μ, λ).

The real Morse lemma classifies assignment germs up to a much coarser right-left equivalence. Conformal realization then equips those germs with examples of AEG metrics. It does not show that all AEG-germ isomorphism classes are represented by the displayed metrics, nor that the metric asymptotics are unique.

Open problem 2.10 (Rigidity beyond conformal realization)*.* Find natural hypotheses derived from arithmetic evaluation under which the local singular AEG germ has a finite classification, and determine which of the Morse and branch models of Section 9 survive that restriction.

3 Multi-zero and logarithmic constructions

3.1 A finite and locally finite construction

The regular-zero theorem is local and does not limit the number of connected components globally. The next model gives a direct answer to the existence part of the multi-zero problem.

Theorem 3.1 (Parallel multi-zero model)**.** *Let $h \in C^\infty(\mathbb{R})$ have only simple zeros. On $M = \mathbb{R}^2$ define*

$$a_h(x, y) = e^x h(y) \tag{5}$$

and

$$g_h = \frac{e^{2x}(h(y)^2 + h'(y)^2)}{\mu^2 + \lambda^2 e^{2x} h(y)^2}(\,\mathrm{d}x^2 + \,\mathrm{d}y^2). \tag{6}$$

Then $(M, g_h, a_h; \mu, \lambda)$ is a regular AES and

$$Z(a_h) = \coprod_{h(c)=0} \mathbb{R} \times \{c\}. \tag{7}$$

In particular, if $h_k(y) = \prod_{j=1}^{k}(y - c_j)$ with distinct c_j, the zero set has exactly k connected components.

Proof. One has

$$\mathrm{d}a_h = e^x h(y)\,\mathrm{d}x + e^x h'(y)\,\mathrm{d}y, \qquad |\,\mathrm{d}a_h|^2_{\text{eucl}} = e^{2x}(h^2 + (h')^2).$$

The simple-zero hypothesis implies that h and h' never vanish simultaneously. The coefficient in (6) is therefore smooth and positive. Theorem 2.2 proves the AEG identity, and e^x never vanishes, so (7) is immediate. □

Remark 3.2 (What the index counts)*.* The polynomial example proves existence of a regular AES with k zero components. It does not prove uniqueness, completeness, homogeneity, or a canonical classification by k. We therefore call it the *parallel k-zero model*, not E_k. The old repository symbol E_k had no stable definition: at different times its subscript suggested construction order, zero count, or singularity count.

Taking $h(y) = \sin y$ gives a countable, locally finite zero family. This example already suggests that countably many visible sheets may be repetitions under a covering group rather than independent components of downstairs geometry.

3.2 A downstairs model and its logarithmic pullback

Let $z = X + iY$ be the standard coordinate on $\mathbb{C}^\times$ and set

$$a_\times(z) = Y, \qquad g_\times = \frac{\mathrm{d}X^2 + \mathrm{d}Y^2}{\mu^2 + \lambda^2 Y^2}. \tag{8}$$

Since $|\,\mathrm{d}Y|^2_{\mathrm{eucl}} = 1$, this is a regular AES. Its zero set is the punctured real axis, hence has two components.

Pull (8) back along the exponential covering

$$\exp : \mathbb{C} \longrightarrow \mathbb{C}^\times, \qquad x + iy \longmapsto e^{x+iy}.$$

The resulting data are

$$\widetilde{a}(x, y) = e^x \sin y, \qquad \widetilde{g} = \frac{e^{2x}}{\mu^2 + \lambda^2 e^{2x} \sin^2 y}(\,\mathrm{d}x^2 + \mathrm{d}y^2), \tag{9}$$

and

$$Z(\widetilde{a}) = \coprod_{k \in \mathbb{Z}} \{y = k\pi\}. \tag{10}$$

Theorem 3.3 (Logarithmic zero lift)**.** *The data in (9) form a regular AES and are invariant under the deck transformation*

$$D(x, y) = (x, y + 2\pi).$$

On the labels in (10), D acts by $k \mapsto k + 2$. Thus the countable upstairs zero family has exactly two deck orbits, corresponding to the positive and negative real half-axes downstairs.

Proof. The formula is the pullback of the regular AES (8); it can also be checked directly from $|\,\mathrm{d}\widetilde{a}|^2_{\mathrm{eucl}} = e^{2x}$. Both $\sin y$ and the metric coefficient are 2π-periodic. Finally, $D(\{y = k\pi\}) = \{y = (k + 2)\pi\}$, proving the orbit statement. $\square$

Remark 3.4 (No singular spine at the missing origin)*.* The coordinate $z = 0$ is absent from $\mathbb{C}^\times$, but the formula $a_\times = Y$ and its metric extend regularly across it. Hence the origin is not a locally essential singularity in the sense of Paper I. The model is a *logarithmic-domain model*, not an essential-singularity model. Calling the missing point a singular spine would erase the local-essentiality condition.

Theorem 3.3 is the rigorous content we retain from the historical $E_{\log}$ intuition:

countably many sheets upstairs $\quad = \quad$ finite downstairs geometry plus deck labels.

No claim is made that it reconstructs every meaning previously attached to $E_{\log}$.

3.3 Sheet labels are not automatically topological states

The label $k \in \mathbb{Z}$ in (10) depends on a choice of fundamental domain. Its parity is invariant under the deck group in this model, but an absolute integer label is not downstairs data. More generally, logarithms of moving complex roots acquire integer shifts whose individual values depend on the chosen lifts; only cycle sums survive the lift gauge. This statement is proved in Section 10 and Appendix B.

There is a second obstruction to interpreting the parallel model as a braid. If $c_1(t) < \cdots < c_k(t)$ are distinct real zero positions throughout a one-parameter family, their order cannot change without a collision. The configuration space of k unordered points on a line has contractible components; it does not carry the Artin braid group. Noncommutative braid monodromy requires motion in a surface, which in this paper comes from complex-valued fields, or else a separately declared transverse probe.

3.4 Metric completeness and provenance limits

The models above are exact solutions of the AEG equation, but completeness is not part of their claim. For example, in the parallel model the conformal factor often decays like e^{2x} as $x \to -\infty$, placing that end at finite metric distance. Completion can add boundary or singular strata and must be audited separately. This is one reason the model record must always list the domain and metric, not only the zero set.

For reference, the construction record is:

Item	Parallel model	Logarithmic model
Domain	$\mathbb{R}^2$	$\mathbb{C} \to \mathbb{C}^\times$
Assignment	$e^x h(y)$	$e^x \sin y = \mathrm{Im}(e^{x+iy})$
Metric	Equation (6)	Equation (9)
Singular set	empty	empty
Zero topology	one line per simple zero of h	countably many lines upstairs, two components downstairs
Global status	regular; completeness not asserted	regular covering model; missing origin removable
Provenance	constructed and proved here	constructed and proved here; historical name used only as motivation

Open problem 3.5 (Natural multi-zero metrics). Classify multi-zero assignments when the metric is fixed by an independent AEG construction, or when completeness, curvature, or expression-history naturality is imposed. Determine whether any canonical index survives those restrictions.

Section 4 gives one nonparallel answer without solving this classification: an automorphic function fixed by a projective arithmetic operator group determines both the metric and a four-valent infinite zero network. The contrast is intentional. The present section proves that multi-zero geometry is analytically abundant; the Hecke construction asks which part of that abundance is forced by arithmetic structure.

4 Arithmetic zero networks and relative divisors

4.1 The projective arithmetic source

Paper I separates literal histories from their induced projective operators. That distinction is decisive here. Over

$$K_4 = \mathbb{Q}(\sqrt{2})$$

the bilateral projective history language contains the two operators

$$\mathsf{T}(\tau) = \tau + \sqrt{2}, \qquad \mathsf{J}(\tau) = -\frac{1}{\tau}, \tag{11}$$

represented by

$$T = \begin{pmatrix} 1 & \sqrt{2} \\ 0 & 1 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Projective evaluation therefore contains the operator subgroup

$$\Gamma_4 = \langle J, T \rangle < \mathrm{PSL}_2(\mathbb{R}), \qquad J^2 = (JT)^4 = 1. \tag{12}$$

It is the Hecke triangle group of signature $(2, 4, \infty)$. Its parabolic translation is T, and the elliptic generators have orders 2 and 4. The associated Hecke–Farey and continued-fraction paths are arithmetic paths in the operator quotient, not distinct merely because they have different word spellings [16, 14].

Let

$$\beta : \mathbb{H} \longrightarrow \mathbb{P}^1(\mathbb{C})$$

be the normalized Γ_4-invariant Hauptmodul. We choose the normalization for which the order-2 elliptic orbit maps to 0, the order-4 elliptic orbit maps to 1, and the cusp orbit maps to ∞. Thus

$$\Gamma_4\backslash\mathbb{H}^* \cong \mathbb{P}^1, \qquad (\mathcal{E}_2, \mathcal{E}_4, \mathcal{C}) \longmapsto (0, 1, \infty), \tag{13}$$

with finite ramification indices 2 and 4 at the elliptic orbits and a cusp over ∞. The existence and normalization are standard for the triangle orbifold [3, 8]. If a cited convention interchanges the two finite branch values, we replace its Hauptmodul by $1 - \beta$; this is the only renormalization being made. In particular, in suitable local coordinates,

$$\begin{aligned} \beta(z) &= c_2 z^2 + O(z^3) && \text{at } \mathcal{E}_2, \\ 1 - \beta(z) &= c_4 z^4 + O(z^5) && \text{at } \mathcal{E}_4, \end{aligned} \tag{14}$$

where $c_2 c_4 \neq 0$. In a quotient cusp coordinate q, the Hauptmodul has a simple pole,

$$\beta(q) = c_\infty q^{-1} + O(1), \qquad c_\infty \neq 0. \tag{15}$$

4.2 The Hecke–Hauptmodul singular AES

The interval $[0, 1]$ is the edge between the two finite orbifold branch values. The next construction turns its full preimage into an exact AEG zero network.

Theorem 4.1 ($q = 4$ Hecke zero network)**.** *Let β have the normalization (13), fix $\mu \neq 0$ and $\lambda \in \mathbb{R}$, and choose a meromorphic square root on $\mathbb{H}$*

$$F^2 = \frac{\beta}{1 - \beta}. \tag{16}$$

Define

$$W = \frac{F - i}{F + i}, \qquad s = \log |W|, \tag{17}$$

and

$$a_4 = \begin{cases} \dfrac{\mu}{\lambda} \sinh(\lambda s), & \lambda \neq 0, \\ \mu s, & \lambda = 0, \end{cases} \qquad g_4 = \left| \frac{\mathrm{d}W}{W} \right|^2 \quad \text{on } \mathbb{H} \setminus \mathcal{E}_4. \tag{18}$$

Then:

1. *$(\mathbb{H}, \mathcal{E}_4, g_4, a_4; \mu, \lambda)$ is a singular AES, and every point of $\mathcal{E}_4$ is locally essential;*
2. *its total zero set is exactly*
$$Z(a_4) = \beta^{-1}([0, 1]); \tag{19}$$
3. *points of $\mathcal{E}_2$ are bivalent regular points of the zero set, while points of $\mathcal{E}_4$ are four-valent singular zeros; the metric completion at each latter point has cone angle 4π;*
4. *after suppressing the bivalent $\mathcal{E}_2$ vertices, $\beta^{-1}([0, 1])$ is the $\{\infty, 4\}$ Hecke zero network: every vertex has valence four and every face runs out to a parabolic cusp and has infinitely many sides;*

5. *the punctured regular metric on $\mathbb{H} \setminus \mathcal{E}_4$ is incomplete at $\mathcal{E}_4$, whose points are at finite distance. Its upstairs completion adds 4π cone points. The induced Γ_4-orbifold length metric, after those points are adjoined, is complete and has its cusp at infinite distance. The underlying coarse quotient has angle π at the image of an order-4 point; none of these statements is smooth geodesic completeness across a regular AES point.*

Proof. The divisor of $\beta/(1-\beta)$ on $\mathbb{H}$ has zeros of order 2 along $\mathcal{E}_2$ and poles of order 4 along $\mathcal{E}_4$. It is therefore even. Since $\mathbb{H}$ is simply connected, it admits the meromorphic square root (16), unique up to sign. The equation $F = \pm i$ would imply $F^2 = -1 = \beta/(1-\beta)$, which is impossible. Hence the Cayley transform W extends holomorphically through the poles of F and is a nowhere-zero finite holomorphic function on $\mathbb{H}$.

Away from $\mathcal{E}_4$, $\mathrm{d}W$ is nonzero: β has no ramification away from the two elliptic orbits, the square root cancels the order-2 ramification at $\mathcal{E}_2$, and the Cayley map has nonzero derivative. Locally choose a logarithm $\zeta = \log W$. Equations (17) and (18) become

$$s = \operatorname{Re}\zeta, \qquad g_4 = |\,\mathrm{d}\zeta|^2.$$

Thus $|\nabla s|_{g_4} = 1$. If $\lambda \neq 0$, then

$$\frac{\mathrm{d}a_4}{\mathrm{d}s} = \mu\cosh(\lambda s), \qquad |\nabla a_4|^2_{g_4} = \mu^2\cosh^2(\lambda s) = \mu^2 + \lambda^2 a_4^2.$$

The calculation for $\lambda = 0$ is immediate.

The function of s defining a_4 is odd and vanishes only at $s = 0$. Moreover, the Cayley transform carries $\mathbb{P}^1(\mathbb{R})$ to the unit circle, so

$$\begin{aligned} a_4 = 0 \iff |W| = 1 \iff F \in \mathbb{P}^1(\mathbb{R}) \\ \iff \frac{\beta}{1-\beta} \in [0,\infty] \iff \beta \in [0,1]. \end{aligned}$$

This proves (19).

At a point of $\mathcal{E}_2$, expansion (14) gives $F = cz + O(z^2)$ and $W = -1 + c'z + O(z^2)$ with $cc' \neq 0$. The zero set is therefore a smooth arc. At a point of $\mathcal{E}_4$, one has $F = cz^{-2}(1 + O(z))$ and hence

$$W = 1 + c'z^2 + O(z^3), \qquad c' \neq 0.$$

Thus $\log W$ has local degree 2. Its real zero has four prongs, and the local metric is the pullback of $|\,\mathrm{d}\zeta|^2$ by a degree-2 branch. The cone-angle calculation in Theorem 2.4 gives 4π. The assignment is critical there, so local essentiality follows from the same $|\nabla a|^2 \geq \mu^2$ obstruction.

The graph $\beta^{-1}([0,1])$ is the orbit of the finite edge in a $(2,4,\infty)$ triangle. Four such edge germs meet at an order-4 vertex; an order-2 point merely subdivides an edge. Suppressing those subdivisions gives the standard $\{\infty,4\}$ Hecke graph. Its complementary regions are the parabolic cusp regions and have infinitely many boundary edges [14].

Finally, Proposition 4.2 below shows that g_4 descends and that $\chi(T) = +1$ for the cusp stabilizer. Thus F and W are single-valued in the quotient cusp coordinate q. The cone calculation places every point of $\mathcal{E}_4$ at finite distance upstairs, with angle 4π. Dividing by its order-4 stabilizer gives angle π on the underlying coarse quotient. At a cusp, (15) gives

$$W = Cq^{\pm 1}(1 + O(q)), \qquad \frac{\mathrm{d}W}{W} = \pm\frac{\mathrm{d}q}{q} + O(\,\mathrm{d}q).$$

The radial integral $\int |\,\mathrm{d}q|/|q|$ diverges. The completed orbifold quotient has a compact core after cusp neighborhoods are removed, so adjoining the finite-distance singular orbit and retaining the infinite cusp end gives the asserted length-metric completeness. The order-2 orbit is smooth upstairs but also has coarse quotient angle π because of its isotropy; its bivalent zero germ distinguishes it from the order-4 singular orbit. □

4.3 Sign-character descent

The square root in (16) is not Γ_4-invariant. This is not an ambiguity to be hidden; it records a natural signed assignment.

Proposition 4.2 (Character-valued descent)**.** *There is a nontrivial character*

$$\chi : \Gamma_4 \longrightarrow \{\pm 1\}$$

with

$$\chi(J) = -1, \qquad \chi(JT) = -1, \qquad \chi(T) = +1, \tag{20}$$

such that

$$F(\gamma\tau) = \chi(\gamma)F(\tau), \quad W(\gamma\tau) = \begin{cases} W(\tau), & \chi(\gamma) = 1, \\ W(\tau)^{-1}, & \chi(\gamma) = -1, \end{cases} \tag{21}$$

and consequently

$$\gamma^* g_4 = g_4, \qquad a_4(\gamma\tau) = \chi(\gamma)a_4(\tau). \tag{22}$$

Thus the metric and zero network descend to the triangle orbifold, while a_4 descends as a section of the real line bundle associated with χ. It becomes an ordinary scalar assignment on the index-two cover associated with $\ker\chi$*.*

Proof. Both $F \circ \gamma$ and F square to the same meromorphic function. Their ratio is therefore a constant in $\{\pm 1\}$ on the connected surface $\mathbb{H}$. Composition makes that sign a character. The order-2 stabilizer J acts by $z \mapsto -z$ in the local coordinate for which $F \sim z$, so $\chi(J) = -1$. At the order-4 orbit, JT acts by $z \mapsto \pm iz$ while $F \sim cz^{-2}$, and hence $\chi(JT) = -1$. Since $\chi(JT) = \chi(J)\chi(T)$, this gives $\chi(T) = +1$ and proves (20). Substitution in the Cayley transform gives (21); logarithmic modulus then sends s to $\chi(\gamma)s$. The metric is unchanged and the odd assignment transforms by the same sign. □

This descent statement is one point at which the signed arithmetic history is strictly richer than the unlabelled zero set. It also fixes the global status of the construction: on the orbifold the natural assignment is line-bundle valued, not a silently single-valued real function.

4.4 From zero histories to relative divisors

The Hecke model supplies an arithmetic source for a zero network, but the phrase "different paths to zero" still has several inequivalent meanings. We use the following audit hierarchy.

Level	Equality being imposed
raw word	literal equality of operation symbols and parentheses
marked history	equality of the sequential tree, accumulator, and operand slots
projective operator	equality after the Paper I evaluation map to $\mathrm{PGL}_2(K)$
endpoint	equality after applying an operator to a declared seed or parameter
prime over K	equality inside one irreducible component defined over the coefficient field K
geometric sheet	equality only after extension to $\overline{K}$ or along a local analytic branch

The first four levels form successively coarser process data. The last two are not further syntactic quotients: they are decompositions of the endpoint equation. Distinct operators can agree at one endpoint, while one K-irreducible component can split into several geometric sheets permuted by Galois or monodromy.

Let B be an integral parameter space over a characteristic-zero field K. If a marked history h and a chosen seed produce a nonzero rational endpoint function $\nu_h \in K(B)^\times$, define its codimension-one zero datum to be the zero part of the principal divisor (ν_h). The exceptional case $\nu_h \equiv 0$ is not a codimension-one divisor: the entire generic base is a zero condition and must be recorded separately. Operator-equivalent histories give the same endpoint function, but the converse need not hold. For a finite set A of operator classes, one possible encoding on a common regular domain is

$$P_A(b, z) = \prod_{[h]\in A} (z - \nu_h(b)). \tag{23}$$

The vanishing condition $P_A(b, 0) = 0$ records that at least one selected operator history reaches zero. Formula (23) is useful but not canonical: it depends on the seed, the finite selection, multiplicities, and the chosen operator quotient.

The following standard algebraic fact identifies the precise irreducible layer once a monic arithmetic polynomial encoding a declared finite history selection has been supplied.

Theorem 4.3 (Arithmetic and geometric zero components)**.** *Let R be a normal finitely generated domain over a characteristic-zero field k, let $B = \operatorname{Spec} R$, let $K = \operatorname{Frac}(R)$, and let*

$$P(z) = z^n + c_1 z^{n-1} + \cdots + c_n \in R[z]$$

have nonzero discriminant. Write

$$\mathcal{D}_P = V(P) \subset \mathbb{A}^1_B.$$

Then:

1. *$\mathcal{D}_P$ is an effective relative Cartier divisor, finite flat of degree n over B;*

2. *the distinct irreducible factorization*

$$P = \prod_{i=1}^{r} P_i \quad \text{in } K[z] \tag{24}$$

   is the unique decomposition of the generic zero divisor into K-prime components; their closures are the horizontal prime components of the associated Weil divisor on $\mathbb{A}^1_B$;

3. *over a separable closure K^{sep}, the roots of each P_i form one transitive $\operatorname{Gal}(K^{\mathrm{sep}}/K)$-orbit, although they may be several geometric sheets;*

4. *on the open set $B^\circ = B \setminus V(\operatorname{disc} P)$, the map $\mathcal{D}_P^\circ \to B^\circ$ is finite étale. If an embedding $k \hookrightarrow \mathbb{C}$ is fixed, set $B^\circ_{\mathbb{C}} = B^\circ \times_k \mathbb{C}$ and $\mathcal{D}^\circ_{P,\mathbb{C}} = \mathcal{D}^\circ_P \times_k \mathbb{C}$. On each connected component of $(B^\circ_{\mathbb{C}})^{\mathrm{an}}$, the analytic covering components of $(\mathcal{D}^\circ_{P,\mathbb{C}})^{\mathrm{an}}$ are exactly the orbits of topological monodromy on one fiber.*

5. *if D_i denotes the horizontal prime closure belonging to P_i and $\widetilde{D}_i$ its normalization, then $\widetilde{D}_i \to B$ is finite and its restriction over B° agrees with the corresponding finite étale covering component.*

Proof. Because P is monic, $R[z]/(P)$ is a free R-module with basis $1, z, \ldots, z^{n-1}$; hence the zero scheme is finite flat of degree n. The element P is a non-zero-divisor in $R[z]$, so its vanishing defines an effective Cartier divisor. Unique factorization in the principal ideal domain $K[z]$ gives (24). Taking closures of the generic prime divisors gives all horizontal codimension-one components; monicity precludes a vertical component. Characteristic zero makes every P_i separable, and irreducibility is equivalent to transitivity of the absolute Galois action on its roots. After inverting the discriminant, P and P' generate the unit ideal in the finite algebra, which is the finite-étale criterion. The monodromy assertion is the standard classification of covering components by their orbits. For item (5), schemes of finite type over a field are excellent, so the normalization of each D_i is finite. Over B°, the prime component is already normal because it is finite étale over the normal base; hence normalization changes nothing there [12, 11]. □

The theorem makes "irreducible paths to zero" field-relative. A K-prime component is an arithmetic unit of zero behavior; its geometric branches need not be individually defined over K. The discriminant records finite collisions and ramification in this monic setting. It does not detect the boundary, metric, domain, or nonproper escape mechanisms included in the structural discriminant of Section 7.

Remark 4.4 (Arithmetic reduction interface). Suppose in addition that the family is given by an integral model over a number ring. Let b be a closed point of the parameter base with finite residue field, lying away from the discriminant and every declared bad-reduction locus. Then the degrees of the irreducible factors of the specialized polynomial P_b are the cycle lengths of geometric Frobenius on the corresponding finite étale fiber. At a discriminant point, the local discriminant records failure of étaleness. After normalizing in a splitting field, inertia records the ramified part, while nonmaximality of an order or a bad integral presentation may contribute additional discriminant factors. Thus discriminant primes, inertia, and reduction factorization are genuinely arithmetic zero data, not merely another drawing of complex monodromy. Using them for AEG histories still presupposes the functorial integral polynomial whose construction is posed below [15].

4.5 What is proved, and what remains a program

For the $q = 4$ model, the arithmetic-to-geometric chain established in this paper is

$$\frac{\text{projective arithmetic histories over } K_4}{\text{operator equality}} \supset \Gamma_4 \longrightarrow \beta^{-1}([0,1]) \longrightarrow (g_4, a_4). \tag{25}$$

If e is one reference edge from an order-2 point to an order-4 point, the edges are translates γe. Equality of two edge labels is equality of the corresponding cosets of the edge stabilizer. Four distinct incident cosets meet at an order-4 endpoint. Raw words representing the same γ have already been identified. Thus the four prongs have an exact operator-coset meaning, but not yet a canonical prime-divisor meaning.

Remark 4.5 (Symmetry is not nonunique factorization). The matrices of Γ_4 are defined over $\mathbb{Z}[\sqrt{2}]$, which is norm-Euclidean and hence a unique factorization domain. The arithmetic richness proved by Theorem 4.1 comes from noncommuting projective operators, Hecke continued-fraction paths, and automorphic uniformization; it does not come from nonunique factorization of elements. If "irreducible paths" is intended specifically to mean inequivalent irreducible factorizations, one must pass to a number ring or order with nontrivial factorization theory, for example a nontrivial class group, and declare whether elements, ideals, units, and order are being quotiented. That extension is not a claim of the present $q = 4$ model.

For completeness, norm-Euclideanity follows directly. Given $x+y\sqrt{2}\in\mathbb{Q}(\sqrt{2})$, choose $a,b\in\mathbb{Z}$ with $|x-a|,|y-b|\le 1/2$. Then

$$\left|N\big((x-a)+(y-b)\sqrt{2}\big)\right| = \left|(x-a)^2-2(y-b)^2\right| \le \frac{1}{2} < 1.$$

Applying this to a quotient gives Euclidean division by the absolute norm.

Open problem 4.6 (History-to-divisor functor). Construct a functorial rule

$$\mathfrak{H} \longmapsto (B_{\mathfrak{H}}, P_{\mathfrak{H}}, \mathcal{D}_{\mathfrak{H}})$$

from a declared category of marked arithmetic histories to parameter spaces, history polynomials, and relative zero divisors, satisfying all of the following:

1. marked-history equivalences and operator equalities act by specified isomorphisms rather than by accidental duplicate factors;
2. ordinary admissibility, projective poles, and chart changes remain visible;
3. composition of histories has a compatible operation on divisors;
4. base change gives the expected Galois and analytic monodromy actions;
5. finite truncations admit a controlled locally finite limit when infinitely many history classes are present.

Determine whether the Hecke zero network is a natural infinite or automorphic instance of this functor, rather than merely its operator-level precursor.

This problem is the main arithmetic naturality gate of the paper. The relative divisor theorem does not solve it: that theorem starts after $P_{\mathfrak{H}}$ has been given. Conversely, the exact Hecke model does not solve it either: it derives a zero network from a projective operator group, but it does not yet associate a finite monic endpoint polynomial to arbitrary AEG histories.

5 The q=4 sign cover and Hecke geodesic knots

The four-pronged zero germ of Theorem 4.1 suggests a knot-theoretic question, but it is not itself a knot. The germ lives in a real surface, and its intersection with a sufficiently small circle consists of four points. A knot appears only after passing to the three-dimensional unit tangent bundle of the global Hecke orbifold. This section makes that passage and keeps the local and global constructions separate.

5.1 The sign cover as an orbifold

Write

$$\mathcal{O}_4=\Gamma_4\backslash\mathbb{H}, \qquad R=JT, \qquad H_4=\ker\chi.$$

Thus

$$\Gamma_4=\langle J,R \mid J^2=R^4=1\rangle, \qquad \chi(J)=\chi(R)=-1, \qquad T=JR\in H_4. \tag{26}$$

The quotient $\mathcal{O}_4^{\chi}=H_4\backslash\mathbb{H}$ is the orbifold on which the assignment a_4 of Proposition 4.2 becomes scalar.

Proposition 5.1 (Signature of the sign cover)**.** *The index-two orbifold cover*

$$\mathcal{O}_4^\chi \longrightarrow \mathcal{O}_4$$

has signature

$$(0; 2; \infty, \infty).$$

More precisely, the order-two point of $\mathcal{O}_4$ has one regular preimage, the order-four point has one preimage of order two, and the cusp has two unramified preimages.

Proof. The relevant stabilizer intersections follow immediately from (26):

$$H_4 \cap \langle J \rangle = 1, \qquad H_4 \cap \langle R \rangle = \langle R^2 \rangle, \qquad \langle T \rangle \subset H_4.$$

Since J and R lie outside H_4, the first two orbifold points each have one preimage. Their new stabilizer orders are respectively 1 and 2. Since the entire parabolic stabilizer lies in H_4, the number of cusp preimages is $[\Gamma_4 : H_4\langle T \rangle] = 2$, and both peripheral covering degrees are one.

It remains only to determine the genus. The orbifold Euler characteristic of the base is

$$\chi_{\mathrm{orb}}(\mathcal{O}_4) = 2 - 1 - \left(1 - \frac{1}{2}\right) - \left(1 - \frac{1}{4}\right) = -\frac{1}{4}.$$

It is multiplied by two under the cover. A genus-g orbifold with one order-two point and two cusps has Euler characteristic

$$2 - 2g - 2 - \left(1 - \frac{1}{2}\right) = -2g - \frac{1}{2}.$$

Equating this with $-1/2$ gives $g = 0$. ☐

Proposition 5.2 (Global scalar-cover uniformization)**.** *Let $\widehat{\mathcal{O}}_4^\chi$ be the coarse AEG surface obtained by adjoining the finite-distance point over the order-four orbit to the regular locus of $H_4\backslash\mathbb{H}$, while leaving its two cusps open. Then W descends to a biholomorphism*

$$W : \widehat{\mathcal{O}}_4^\chi \longrightarrow \mathbb{C}^\times. \tag{27}$$

Under this identification,

$$g_4 = \left|\frac{\mathrm{d}W}{W}\right|^2$$

is the complete flat cylindrical metric and the scalar zero set is the single circle

$$Z(a_4)/H_4 = \{|W| = 1\}.$$

Proof. Proposition 5.1 identifies the compactified coarse surface with $\mathbb{P}^1$, with two punctures coming from the cusps. The H_4-invariance of F makes W single-valued on this surface. It has neither a zero nor a pole in the interior. At the two cusps the expansion in the proof of Theorem 4.1 is

$$W = Cq^{\pm 1}(1 + O(q)).$$

The two signs are opposite: an element of $\Gamma_4 \setminus H_4$ interchanges the two cusp lifts and sends W to W^{-1}. Hence the meromorphic extension of W to the compact coarse sphere has one simple zero and one simple pole, so it has degree one.

At the remaining order-two orbifold point of the cover, an upstairs coordinate z satisfies $W = 1 + cz^2 + O(z^4)$, with $c \neq 0$. Thus in the coarse coordinate z^2, W is unramified. It is also unramified at the regular preimage of the old order-two point and everywhere else. This proves (27). The metric and zero-circle statements now follow directly, and completeness is the divergence of $\int |\,\mathrm{d}W|/|W|$ at 0 and ∞. ☐

Remark 5.3 (The zero circle is not a hyperbolic periodic orbit)*.* The circle in Proposition 5.2 is a zero level and a geodesic for the flat AEG metric. No claim is made that it is a closed geodesic for the original hyperbolic orbifold metric. The periodic-orbit knots below instead come from unit-tangent lifts of hyperbolic axes. In particular, Proposition 5.2 forgets the residual order-two orbifold isotropy in the coarse coordinate, whereas $T^1\mathcal{O}_4^\chi$ below retains that isotropy. The flat unit tangent bundle of the coarse cylinder is not being identified with the Hecke unit tangent bundle.

5.2 Hyperbolic compactification and the cusp link

Let

$$M_4 = T^1\mathcal{O}_4, \qquad M_4^\chi = T^1\mathcal{O}_4^\chi.$$

The unit tangent bundle of an orientable hyperbolic orbifold is a genuine oriented three-manifold: the rotations at an orbifold point act freely on unit tangent vectors.

Definition 5.4 (Hyperbolic cusp compactification)**.** For a cusp of an oriented hyperbolic two-orbifold, the *hyperbolic compactification* of its unit tangent bundle is the Dehn filling in which the circle of vectors tangent to a horocycle bounds a meridian disc. The core of the attached solid torus is denoted by K_∞. This convention is part of the definition; changing the filling slope can change the ambient lens space.

Theorem 5.5 (The $q = 4$ cusp knot and its sign-cover link)**.** *With the preceding compactification convention:*

1. *the compactification of M_4 is*

$$\overline{M}_4 \cong L(2,1) \cong \mathbb{RP}^3, \qquad M_4 \cong \mathbb{RP}^3 \setminus K_\infty, \tag{28}$$

   and K_∞ is null-homologous in $\mathbb{RP}^3$;

2. *the covering $M_4^\chi \to M_4$ extends over the cusp fillings to the universal double cover*

$$S^3 \longrightarrow \mathbb{RP}^3;$$

3. *the full inverse image of K_∞ is the two-component torus link $T(2,4)$, and hence*

$$M_4^\chi \cong S^3 \setminus T(2,4). \tag{29}$$

   The two components have absolute mutual linking number 2*.*

Proof. For the triangle orbifold of signature (p, q, ∞), the standard unit-tangent bundle calculation gives

$$\overline{T^1\mathcal{O}_{p,q}} \cong L(pq - p - q, p - 1). \tag{30}$$

In the same calculation the added cusp fiber lies on the median Heegaard torus. If (c_P, c_Q) is the basis used there, the two meridians and the cusp fiber have classes

$$m_P = (p - 1, -1), \qquad m_Q = (-1, q - 1), \qquad a_\infty = (1, 1). \tag{31}$$

These statements are the lens-space and slope computation of [6, Lemmas 5.2 and 5.3]. Setting $p = 2$ and $q = 4$ gives $L(2,1) \cong \mathbb{RP}^3$, and deleting the added core recovers M_4. Moreover, for these values,

$$a_\infty = 2m_P + m_Q.$$

Both meridians vanish after the two solid tori are attached, so $[K_\infty] = 0$ in $H_1(\mathbb{RP}^3; \mathbb{Z})$.

By Proposition 5.1, each cusp has two lifts and each peripheral covering degree is one. More explicitly, both the parabolic class T and the central unit-tangent fiber class lie in the covering subgroup. Therefore every peripheral torus maps homeomorphically to the base peripheral torus and the chosen meridian slope lifts unchanged. The double cover consequently extends, without branching, over the two solid tori used for hyperbolic filling. The extended cover is connected, so it is the universal double cover $S^3 \to \mathbb{RP}^3$.

It remains to identify the lifted link; this is where the non-coprime pair $(2, 4)$ requires an explicit slope check. For $p = 2, q = 4$, equations (31) read

$$m_P = (1, -1), \qquad m_Q = (-1, 3), \qquad a_\infty = (1, 1).$$

The compactified first homology is $\mathbb{Z}^2/\langle m_P, m_Q\rangle$, whose order is $|\det(m_P, m_Q)| = 2$. The homomorphism $(x, y) \mapsto x + y \pmod 2$ kills both meridians, so it is the restriction to the median torus of the quotient map to $H_1(\mathbb{RP}^3; \mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$. Therefore the universal cover of the median torus corresponds to the index-two lattice

$$\Lambda = \{(x, y) \in \mathbb{Z}^2 : x + y \equiv 0 \pmod 2\}.$$

The pair (a_∞, m_P) is a basis of Λ, and in that basis

$$m_P = (0, 1), \qquad m_Q = (1, -2), \qquad a_\infty = (1, 0).$$

Thus the lifted Heegaard splitting is the standard genus-one splitting of S^3. Moreover $a_\infty = m_Q + 2m_P$. Since a_∞ is trivial in the deck quotient, its inverse image has two components, each mapping homeomorphically to K_∞. On the lifted Heegaard torus these are two parallel copies of the primitive slope $2m_P + m_Q$. They form $T(4, 2)$, which is isotopic to $T(2, 4)$. The usual torus framing gives absolute mutual linking number 2; its sign depends on the orientation convention. Removing the link proves (29). □

Warning 5.6 (What has and has not produced the link). Theorem 5.5 is global. It uses the $(2, 4, \infty)$ orbifold, its unit tangent bundle, the character cover, and a specified cusp filling. It does not identify the small link of the real four-pronged surface germ with $T(2, 4)$. Nor can one obtain the conclusion by substituting $p = 2, q = 4$ into an S^3 torus-*knot* theorem that assumes p and q coprime. In particular, the central-extension presentation $\langle x, y \mid x^2 = y^4\rangle$ has abelianization $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, whereas the complement of a two-component link in S^3 has first homology $\mathbb{Z}^2$. The passage to the index-two cover in Theorem 5.5 is essential.

5.3 Primitive Hecke classes and geodesic knots

Definition 5.7 (Oriented Hecke geodesic knot)**.** Let $\gamma \in \Gamma_4$ be primitive and hyperbolic. Its invariant axis $A_\gamma \subset \mathbb{H}$ is oriented from the repelling to the attracting fixed point. The corresponding closed geodesic in $\mathcal{O}_4$ is denoted C_γ. Its unit-tangent lift is

$$K_\gamma = \{(x, \dot{C}_\gamma(x)) : x \in C_\gamma\} \subset M_4. \tag{32}$$

Proposition 5.8 (Arithmetic class to periodic-orbit knot)**.** *Primitive hyperbolic conjugacy classes in Γ_4 are in bijection with primitive oriented periodic orbits of the geodesic flow on M_4. Under this correspondence K_γ is an embedded oriented circle. Replacing γ by γ^{-1} reverses the orientation of the base geodesic and in general gives the oppositely directed periodic orbit rather than the same oriented knot.*

If

$$\widetilde{\gamma} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Q}(\sqrt{2}))$$

is a lift, then its fixed points satisfy

$$cx^2 + (d-a)x - b = 0, \qquad \Delta_\gamma = (\operatorname{tr}\widetilde{\gamma})^2 - 4, \tag{33}$$

and the geodesic length is

$$\ell(K_\gamma) = 2 \operatorname{arcosh} \frac{|\operatorname{tr}\widetilde{\gamma}|}{2}. \tag{34}$$

Thus the knot is canonically labelled by quadratic Hecke data, although its ambient isotopy type need not be determined by the trace or discriminant alone.

Proof. A hyperbolic element acts by translation along its oriented axis. Taking the axis modulo the cyclic subgroup it generates gives a closed geodesic; conjugation only changes its lift in $\mathbb{H}$. Conversely, lifting an oriented closed geodesic to $\mathbb{H}$ gives an axis and its primitive deck translation. Primitivity removes multiple traversal. A primitive periodic orbit of a nonsingular flow is an embedded circle, even when its projection to the base orbifold self-intersects. The fixed-point equation follows by solving $x = (ax+b)/(cx+d)$, and the trace–length formula is the standard eigenvalue calculation for a hyperbolic element of $\mathrm{PSL}_2(\mathbb{R})$. □

The sign character has an immediate three-dimensional meaning for these knots.

Proposition 5.9 (Sign-controlled lifting)**.** *Let $\pi : M_4^\chi \to M_4$ be the sign cover and let K_γ be as above. Then*

$$\pi^{-1}(K_\gamma) \quad \textit{has} \quad \begin{cases} 2 \textit{ components, each of degree } 1, & \chi(\gamma) = +1, \\ 1 \textit{ component of degree } 2, & \chi(\gamma) = -1. \end{cases}$$

In the second case the lifted periodic orbit is represented by $\gamma^2 \in H_4$.

Proof. The monodromy of the double cover around the periodic orbit is the image of its projected deck transformation in $\Gamma_4/H_4 \cong \{\pm 1\}$, namely $\chi(\gamma)$. The two alternatives are the elementary lifting alternatives for a connected double cover. If the monodromy is nontrivial, traversing the base orbit twice closes the lift, and the corresponding element is γ^2. □

5.4 The classical cutting code and a linking formula

We next record the precise part of the arithmetic–knot relation that is already available from the classical Hecke tessellation. Choose the standard adapted $(2, 4, \infty)$ tessellation and the orientations used in [6]. A periodic geodesic has a reduced cyclic cutting code

$$\left(uv^{j_1} uv^{j_2} \cdots uv^{j_m}\right)^{\mathbb{Z}}, \qquad j_k \in \{1, 2, 3\}, \tag{35}$$

unique up to cyclic permutation. Because K_∞ is null-homologous by Theorem 5.5, its linking number with any disjoint oriented knot is defined using an integral Seifert surface and is integer-valued.

Theorem 5.10 (The $q = 4$ cusp-linking formula)**.** *For a periodic orbit K_γ with reduced code (35), the integer linking number in $\overline{M}_4 \cong \mathbb{RP}^3$ is*

$$\mathrm{Lk}_{\overline{M}_4}(K_\gamma, K_\infty) = \sum_{k=1}^{m} (j_k - 2) = \#\{k : j_k = 3\} - \#\{k : j_k = 1\}. \tag{36}$$

The sign is tied to the displayed orientation convention; reversing one component reverses the sign.

Proof. For a (p,q,∞) orbifold, define the wheel turn of a reduced code $u^{i_1}v^{j_1}\cdots u^{i_m}v^{j_m}$ by

$$\Theta_{\mathrm{wheel}} = \sum_{k=1}^{m}\left(\frac{i_k - p/2}{p} + \frac{j_k - q/2}{q}\right).$$

The classical cusp-linking theorem gives

$$\mathrm{Lk}(K_\gamma, K_\infty) = \frac{pq}{pq-p-q}\,\Theta_{\mathrm{wheel}} \qquad \text{[6, Proposition 5.7]}.$$

For $p=2$, every i_k equals 1, so its contribution vanishes. For $q=4$, the remaining contribution is $\Theta_{\mathrm{wheel}} = \frac{1}{4}\sum_k(j_k-2)$, while $pq/(pq-p-q)=4$. Substitution proves (36). □

Corollary 5.11 (Parity of linking and splitting in the sign cover)**.** *For the geometric cutting code* (35)*, the standard turn-to-deck-transformation calibration gives*

$$\chi(\gamma) = (-1)^{m+\sum_k j_k} = (-1)^{m+\mathrm{Lk}(K_\gamma,K_\infty)}. \tag{37}$$

Consequently, the full inverse image of K_γ in $S^3\setminus T(2,4)$ has two components when $m+\mathrm{Lk}(K_\gamma,K_\infty)$ is even and one component when it is odd.

Proof. Every u-turn is the nontrivial rotation at an order-two vertex, hence a conjugate of J. Every v^{j_k}-turn is, in the moving vertex frame, a conjugate of R^{j_k} or R^{-j_k}, according to the clockwise orientation convention. Because χ is conjugacy invariant and $\chi(R^{\pm j_k}) = (-1)^{j_k}$, multiplying the turn contributions gives $\chi(\gamma) = \prod_k(-1)^{1+j_k}$. This argument uses the classical geometric code-to-group calibration; it does not identify the code with a literal Paper I history word. On the other hand, Theorem 5.10 gives

$$\mathrm{Lk}(K_\gamma,K_\infty) = \sum_k j_k - 2m \equiv \sum_k j_k \pmod 2.$$

The last assertion follows from Proposition 5.9. □

Theorem 5.12 (The zero dessin as the classical coding spine)**.** *Match the fundamental triangle used to normalize β with the triangle used in the classical $(2,4,\infty)$ cutting construction. Then:*

1. *the Dehornoy–Pinsky graph $G[2,4,\infty]$ is exactly the AEG zero dessin,*

$$G[2,4,\infty] = \beta^{-1}([0,1]) = Z(a_4), \tag{38}$$

   as an embedded Γ_4-graph in $\mathbb{H}$;

2. *this graph is the barycentric subdivision of the four-regular tree dual to the adapted tessellation by ideal quadrilaterals; suppressing its bivalent order-two vertices recovers that dual tree;*

3. *the two endpoints of a hyperbolic axis determine the unique Dehornoy–Pinsky admissible bi-infinite path when $r=\infty$. For an axis stabilized by γ, this path is periodic modulo γ; after suppressing the bivalent vertices it is a no-backtracking path in the dual zero tree, and its turns give the cyclic code* (35)*;*

4. *replacing the admissible graph path by the classical roundabout path and then thickening the allowed visual-direction intervals produces the template. The unit-tangent knot K_γ is isotopic to the corresponding template orbit, simultaneously for every finite collection of periodic orbits.*

Proof. The graph $G[2,4,\infty]$ is defined as the Γ_4-orbit of the geodesic segment joining the order-two and order-four vertices. The proof of Theorem 4.1 identifies $\beta^{-1}([0,1])$ with the orbit of the same segment, which proves (38). Suppressing the order-two midpoint of each pair of adjacent ideal quadrilaterals gives their ordinary dual tree.

The remaining assertions are the classical coding and template theorems. For $r = \infty$ the coding graph is a tree and the spectacles choice is unique. It assigns to the two ideal endpoints of every geodesic an admissible graph path, including the boundary cases for which a naive crossed-tile itinerary would be ambiguous. Periodicity of the geodesic is equivalent to periodicity of the reduced path code. The classical roundabout replacement records the entry and exit data inside each tile; retaining the corresponding interval of unit tangent directions produces the ribbons. The tearing construction then gives the asserted simultaneous isotopy [6, 7]. □

Warning 5.13 (The spine is not the template). The hyperbolic axis is not literally contained in $Z(a_4)$. Its tile itinerary is represented on the dual zero tree. Moreover, the bare tree does not remember the visual direction needed to select the chord and the over–under placement of ribbons in the unit tangent bundle. Thus Theorem 5.12 does not identify an AEG zero tube with the classical template; it identifies the AEG zero dessin with the one-dimensional coding spine from which the directed template is constructed.

Proposition 5.14 (The operator-quotient history-to-knot map)**.** *Let* $\mathrm{Hist}_4^{\mathrm{ph}}$ *be the Paper I words in* $\mathtt{t}_+, \mathtt{t}_-$*, and* $\mathtt{j}$ *whose projective operator is primitive and hyperbolic. Then*

$$\mathcal{K}_{\mathrm{op}} : \mathrm{Hist}_4^{\mathrm{ph}} \longrightarrow \{\textit{oriented primitive geodesic-flow periodic orbits in } M_4\}, \qquad h \longmapsto K_{\rho(h)}, \quad (39)$$

is surjective and has the following invariance properties.

1. *A cyclic shift of the chronological word does not change the oriented knot: its operator is conjugate to* $\rho(h)$*.*
2. *Reversing the word and inverting every letter replaces the orbit by the time-reversed orbit* $K_{\rho(h)^{-1}}$*, which in general is not merely the same embedded circle with its orientation reversed.*
3. *The map factors through the conjugacy class of the projective operator. In particular, inserting Paper I's neutral eight-letter word* ω_4*, or making any other marked-history change that preserves that conjugacy class, cannot be detected by the unframed knot.*

No ordinary real-input admissibility is asserted by (39)*.*

Proof. Write a chronological word as $h = (g_1, \dots, g_n)$, with matrix representatives M_i. Paper I's convention gives $\rho(h) = [M_n \cdots M_1]$. The cyclic shift $(g_2, \dots, g_n, g_1)$ has operator

$$[M_1 M_n \cdots M_2] = [M_1]\rho(h)[M_1]^{-1},$$

so Proposition 5.8 proves the first claim. The path inverse evaluates to the inverse projective operator, proving the second. The third is then immediate, and $\rho(\omega_4) = 1$ by Paper I. Finally, every element of Γ_4 is represented by this history alphabet; hence every primitive hyperbolic class, and therefore every knot in the target, has a preimage. □

Open problem 5.15 (Enhancing operator histories beyond the knot quotient)*.* Proposition 5.14 settles the unadorned operator-level map but also exhibits its large kernel. Construct a declared marked cyclic equivalence and a canonical enhancement

$$\mathrm{Hist}_4^{\mathrm{ph}} /{\sim_{\mathrm{cyc}}} \longrightarrow \{\text{classical cyclic cutting codes with AEG history decoration}\}$$

that specifies ordinary-arithmetic admissibility, base edge, orientation, and the residual data carried by histories with conjugate operators. It must prove compatibility with the cutting construction of Theorem 5.12 and characterize the marked and syntactic residual structure inside the conjugacy-class fibers of $\mathcal{K}_{\mathrm{op}}$, or the still larger fibers after forgetting to ambient isotopy type. Without such an enhancement, the classical code is an intrinsic code of the Hecke operator class, not of the full marked AEG history.

Remark 5.16 (Current novelty boundary). Theorems 5.5 and 5.10 give a rigorous route

$$\begin{aligned}\text{Hecke conjugacy and cutting data} &\longrightarrow \text{closed geodesics}\\ &\longrightarrow \text{knots and linking numbers.}\end{aligned}$$

The character χ additionally controls how the knots split in the natural S^3 cover. The classical geodesic-flow and linking parts are not new knot invariants. The AEG-specific frontier is to enhance Proposition 5.14 as required by Problem 5.15, and then determine whether the assignment, cone, sign, divisor, or history data define an isotopy invariant not already determined by the classical Hecke code or its known Rademacher-type functions. Nothing in this section asserts such a separation theorem.

6 Register-derived finite covers and history naturality

6.1 A quadratic register model for history naturality

The relative-divisor theorem starts with a polynomial already supplied. The following restricted model runs in the other direction: a declared algebraic register and a projective arithmetic operator force the polynomial by elimination. It is deliberately smaller than Problem 4.6. In particular, the divisor below records the possible *endpoint values* of a register; it is not, by itself, the divisor of histories that reach zero, and its two sheets are not two inequivalent marked histories.

Put

$$k = \mathbb{Q}(\sqrt{2}), \qquad R = k[t, t^{-1}], \qquad B = \operatorname{Spec} R,$$

and supply the quadratic register cover

$$\pi : \widetilde{B} = \operatorname{Spec} k[u, u^{-1}] \longrightarrow B, \qquad t = u^2. \tag{40}$$

This cover is extra typed input: extracting a square root is not one of the four elementary contexts of Paper I. Let $\gamma \in \mathrm{Hist}_4$ be a marked history in the distinguished Hecke sublanguage and set $h = \rho(\gamma) \in \Gamma_4$. Choose a determinant-one representative

$$M_h = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}[\sqrt{2}]).$$

The endpoint part of the computation graph is the B-morphism

$$\iota_h : \widetilde{B} \longrightarrow \mathbb{P}^1_B, \qquad u \longmapsto [au + b : cu + d]. \tag{41}$$

We call the triple $(\widetilde{B}, u, \iota_h)$ the *register-decorated endpoint correspondence*, and denote its scheme-theoretic image by D_h.

Theorem 6.1 (Forced projective quadratic). *With the notation above, the following statements hold.*

1. *The endpoint divisor is cut out in $\mathbb{P}^1_B$, with homogeneous coordinates $[Z:W]$, by the binary quadratic*
$$\mathcal{P}_h(Z,W;t) = (dZ - bW)^2 - t(aW - cZ)^2. \tag{42}$$
It is independent of the scalar representative of h, up to multiplication by a unit of R.

2. *The morphism ι_h identifies $\widetilde{B}$ with D_h. Consequently $D_h \to B$ is an integral finite étale cover of degree two, hence finite flat.*

3. *On the chart $W = 1$, its equation is*
$$P_h(z,t) = (dz - b)^2 - t(a - cz)^2. \tag{43}$$
As a polynomial in z over $k(t)$, it is quadratic and irreducible, and
$$\operatorname{disc}_z(P_h) = 4t(ad - bc)^2 = 4t. \tag{44}$$
Its leading coefficient is $d^2 - tc^2$, which need not be a unit of R. Thus the affine equation must not be inserted directly into the monic affine relative-divisor theorem.

4. *After the nonconstant Kummer base change $\widetilde{B} \to B$, the divisor splits into two disjoint sections:*
$$\begin{aligned}\mathcal{P}_h = {} & \big(dZ - bW - u(aW - cZ)\big) \\ & \cdot \big(dZ - bW + u(aW - cZ)\big),\end{aligned} \tag{45}$$
whose values are respectively $h(u)$ and $h(-u)$. Extension of the constant field from k to $\overline{k}$ does not split the generic polynomial: t is still nonsquare in $\overline{k}(t)$.

5. *If $g \in \Gamma_4$, then*
$$D_{g \circ h} = g_{B*} D_h. \tag{46}$$
More explicitly, for $M_g = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$,
$$\mathcal{P}_{g \circ h}(Z,W;t) = \mathcal{P}_h(sZ - qW, pW - rZ; t). \tag{47}$$
Hence, for chronological concatenation—first γ, then δ—
$$D_{\gamma\delta} = \rho(\delta)_{B*} D_\gamma, \qquad \rho(\gamma\delta) = \rho(\delta)\rho(\gamma). \tag{48}$$

Proof. Write $\Delta = ad - bc$. At the projective point $[Z:W] = [au + b : cu + d]$, the two rows of the adjugate matrix give
$$dZ - bW = \Delta u, \qquad aW - cZ = \Delta.$$
Eliminating u from these equations and $u^2 = t$ gives (42). Scaling M_h by $q \in k^\times$ scales $\mathcal{P}_h$ by q^2, so its Cartier divisor is unchanged.

For the identity operator, the divisor is
$$D_1 = V(Z^2 - tW^2).$$
It misses $W = 0$, and on $W = 1$ its coordinate ring is
$$R[z]/(z^2 - t) \cong k[u, u^{-1}].$$

It is therefore integral and finite free of rank two over R. Since $4t \in R^\times$, it is also étale. The automorphism h_B of $\mathbb{P}^1_B$ carries D_1 to D_h, proving the first two claims.

Expanding (43) gives coefficients

$$A = d^2 - tc^2, \qquad B = 2(tac - bd), \qquad C = b^2 - ta^2.$$

The leading coefficient is nonzero in $k(t)$, since otherwise the transcendence of t would force $c = d = 0$. Direct calculation yields $B^2 - 4AC = 4t\Delta^2$. The polynomial $z^2 - t$ is irreducible over $k(t)$, and an invertible projective change of coordinates preserves irreducibility, proving the generic assertion. Factoring after $t = u^2$ gives (45). The two factors are disjoint because $2u$ is a unit. The same nonsquare argument over $\overline{k}(t)$ proves the constant-field qualification.

Finally, $D_h = h_B(D_1)$, so $D_{g\circ h} = g_B(D_h)$. Substituting the inverse coordinates $[sZ - qW : -rZ + pW]$ gives (47). The chronological rule is precisely the convention of Paper I. These are elementary instances of finite-flat and finite-étale base change [12, 11]. □

The geometric generic fiber has two points over $\overline{k(t)}$, but the total cover remains integral after extending only the constants to $\overline{k}$. Equation (45) is instead a splitting after the nonconstant register cover itself.

Remark 6.2 (The affine pole is not a collision). The projective endpoint is finite on a chosen register branch precisely when $cu + d \neq 0$. Requiring both conjugate branches to be finite removes

$$V\big(\mathrm{N}_{k(u)/k(t)}(cu + d)\big) = V(d^2 - c^2 t)$$

from B. If $c, d \neq 0$, then at $t = (d/c)^2$ one endpoint moves to infinity and the affine polynomial loses degree. Its discriminant remains nonzero, so no two projective roots collide. The homogeneous divisor is still finite étale there.

If instead one compactifies the base to $\overline{B} = \operatorname{Spec} k[t]$, the same projective equation remains integral and finite flat, but it is ramified at $t = 0$. Its fiber there is a nonreduced double point, and the ramification is tame of index two. Thus the $\mathbb{G}_m$-model has Frobenius but no inertia, whereas this boundary compactification supplies the single Kummer inertia point.

6.2 Time-tagged traces, zero hits, and ordinary poles

The endpoint divisor forgets the intermediate computation. There is, however, a restricted decorated object which is still forced by the same marked history. For $\gamma = (g_1, \ldots, g_n)$, write

$$h_0 = 1, \qquad h_j = \rho(g_1, \ldots, g_j) = g_j \circ \cdots \circ g_1.$$

Define the *time-tagged register trace*

$$\mathcal{T}_\gamma = \bigsqcup_{j=0}^{n} (D_{h_j} \times \{j\}), \tag{49}$$

together with the transition label g_{j+1} from time j to time $j+1$. The tags are part of the object; forgetting them merely superposes endpoint divisors and destroys chronology.

For ordinary arithmetic, define first a divisor on the projective input line. Every letter $\mathsf{j}[x] = -1/x$ requests division by the current value. Put

$$\Pi_\gamma = \sum_{\{j :\, g_{j+1} = \mathsf{j}\}} h_j^*[0] \quad \text{on } \mathbb{P}^1_k, \tag{50}$$

Let $\Pi_{\gamma,B} = \Pi_\gamma \times_k B$ on $\mathbb{P}^1_B$, and let $\Pi_\gamma^{(u)} = \iota_1^* \Pi_{\gamma,B}$ be its pullback to the register cover. Multiplicity records repeated requests at the same input.

Proposition 6.3 (Restricted trace composition and pole cocycle)**.** *For $q = 4$ histories the following hold.*

1. *Chronological concatenation appends the trace of δ, evaluated from the terminal state of γ, to the tagged trace of γ. The transition labels and time reindexing determine this operation uniquely.*

2. *The ordinary pole divisors satisfy the exact cocycle identity*

$$\Pi_{\gamma\delta} = \Pi_\gamma + \rho(\gamma)^*\Pi_\delta. \tag{51}$$

   The branchwise ordinary-admissible locus is $\widetilde{B} \setminus \mathrm{Supp}(\Pi_\gamma^{(u)})$.

3. *If $M_{h_j} = \left(\begin{smallmatrix} a_j & b_j \\ c_j & d_j \end{smallmatrix}\right)$, then the parameters at which time j reaches zero are the intersection of D_{h_j} with the zero section $[0:1]$, hence are cut out on B by*

$$b_j^2 - ta_j^2 = 0. \tag{52}$$

   On $B = \mathbb{G}_{m,k}$ this locus is empty when $a_jb_j = 0$, and otherwise is the single divisor $t = (b_j/a_j)^2$.

Proof. The prefix operators of $\gamma\delta$ are first those of γ, and then the maps $q_j \circ \rho(\gamma)$, where q_j runs through the prefixes of δ. This proves the trace statement. A pole occurring in the second block has input divisor $\rho(\gamma)^*(q_j^*[0])$; summing the first and second blocks gives (51). Since translations have no finite pole and every inversion pole is listed, the complement of the pulled-back support is exactly the ordinary-admissible locus for a chosen register branch. Finally, substitution of $[Z : W] = [0 : 1]$ in (42) gives (52). □

The proposition supplies a genuine composition law for a *restricted, decorated correspondence*: the quadratic seed cover is fixed in advance, every time is tagged, and every transition and pole request is retained. It is not a functor from arbitrary marked histories to prime divisors. In particular, $\mathcal{T}_\gamma$ is generally a disconnected multisection, while $D_{\rho(\gamma)}$ is only its terminal prime divisor.

Example 6.4 (A neutral operator with nonneutral trace)**.** Let $\lambda = \sqrt{2}$, and let

$$\omega_4 = (\mathsf{t}_+, \mathsf{j}, \mathsf{t}_+, \mathsf{j}, \mathsf{t}_+, \mathsf{j}, \mathsf{t}_+, \mathsf{j}).$$

With the chronological convention of Paper I, $\rho(\omega_4) = (JT)^4 = 1$. Hence $D_{\omega_4} = D_1$, and even the register-decorated terminal map is the same as for the empty history. The four universal inversion requests occur at

$$x = -\sqrt{2}, \qquad x = -1/\sqrt{2}, \qquad x = 0, \qquad x = \infty.$$

The register $u \in \mathbb{G}_m$ omits the last two, so

$$\Pi_{\omega_4}^{(u)} = [u = -\sqrt{2}] + [u = -1/\sqrt{2}], \qquad \Pi_\varnothing^{(u)} = 0. \tag{53}$$

The maximal base open on which both conjugate register branches are ordinarily admissible therefore removes $t = 2$ and $t = 1/2$. The tagged traces are also different: one has nine time slices and the other only its initial slice. Thus endpoint equality does not imply equality of ordinary domains or computation traces.

6.3 What the divisor retains, and what it necessarily forgets

Proposition 6.5 (Exact collapse levels in the quadratic model)**.** *Let $h_1, h_2 \in \Gamma_4$.*

1. *As abstract covers of B, all $D_h \to B$ are isomorphic to the same Kummer cover $\widetilde{B} \to B$. Their monodromy, ramification behavior, and Frobenius cycle types therefore do not depend on h. The operator enters only through the embedding into the fixed output line $\mathbb{P}^1_B$.*

2. *Let $\epsilon(z) = -z$. Then*

$$D_{h_1} = D_{h_2} \iff h_2^{-1}h_1 \in \{1, \epsilon\} \quad \text{in } \mathrm{PGL}_2(k). \tag{54}$$

   Since ϵ sends the upper half-plane to the lower half-plane, $\Gamma_4 \cap \{1, \epsilon\} = \{1\}$. Thus the embedded divisor recovers the operator inside Γ_4, although the abstract cover does not.

3. *Equality of register-decorated endpoint maps $\iota_{h_1} = \iota_{h_2}$ implies $h_1 = h_2$. Nevertheless the assignment $\gamma \mapsto (\widetilde{B}, u, \iota_{\rho(\gamma)})$ factors through ρ. Every pair of distinct histories with the same projective operator, including ω_4 and the empty word, has the same terminal correspondence.*

4. *At a single chosen geometric register value u_0, endpoint equality is coarser:*

$$h_1(u_0) = h_2(u_0) \iff h_2^{-1}h_1 \in \mathrm{Stab}_{\Gamma_4}(u_0). \tag{55}$$

   By contrast, equality at the generic register u forces operator equality.

Proof. The first assertion follows from $D_h = h_B(D_1)$. A constant projective automorphism preserving D_1 restricts to a B-automorphism of the quadratic cover. That cover has only the identity and the deck involution $u \mapsto -u$. Because u is transcendental over k, the corresponding rational maps are respectively $z \mapsto z$ and $z \mapsto -z$. This proves (54). The orientation statement gives its intersection with Γ_4. If $h_1(u) = h_2(u)$ in $k(u)$, two fractional linear maps agree as rational functions and hence define the same projective operator. The final assertion is the usual orbit–stabilizer criterion. □

This proposition is the plain-divisor no-go in its precise form. Forgetting the embedding collapses all operators. Keeping the embedding can recover the restricted Γ_4 operator, but not a marked word in its evaluation fiber. Keeping the register distinguishes the two Kummer lifts; keeping the tagged prefix trace and pole divisor additionally retains intermediate states and ordinary domain failures. None of these statements proves that the trace is a complete invariant of literal or marked-history equality.

6.4 Frobenius in the quadratic model

Let $\mathcal{O}_k = \mathbb{Z}[\sqrt{2}]$. Because the standard generators of Γ_4 lie in $\mathrm{SL}_2(\mathcal{O}_k)$, the preceding construction has an integral model over $\mathcal{O}_k[1/2, t, t^{-1}]$.

Proposition 6.6 (Quadratic Frobenius cycle)**.** *Let b be a closed point of this integral base, let $\kappa(b) = \mathbb{F}_q$, and write $t_b \in \mathbb{F}_q^\times$ for the value of the register parameter. On the two geometric endpoint values of D_h, either arithmetic or geometric Frobenius has cycle type*

$$\begin{cases} (1)(1), & t_b \text{ is a square in } \mathbb{F}_q, \\ (12), & t_b \text{ is a nonsquare in } \mathbb{F}_q. \end{cases} \tag{56}$$

This cycle type is independent of h; the embedding merely sends $\{\pm\sqrt{t_b}\}$ to $\{h(\pm\sqrt{t_b})\}$.

Proof. For a root $r^2 = t_b$ in $\overline{\mathbb{F}}_q$, one has $r^q = t_b^{(q-1)/2} r$. Thus Frobenius fixes the two signs when the quadratic character is $+1$ and interchanges them when it is -1. In a two-element permutation group arithmetic and geometric Frobenius have the same cycle type. The coefficients of h lie in the residue field, so Frobenius commutes with the projective coordinate change [15]. $\square$

For example, at the prime $(7, \sqrt{2} - 3)$ the residue field is $\mathbb{F}_7$. The specialization $t = 2$ splits because $2 = 3^2$, whereas $t = 3$ gives a transposition. This arithmetic cycle belongs to the supplied register cover, not to a new history invariant.

6.5 A quartic test separating constant and geometric components

The quadratic model is arithmetically prime and remains integral after every extension of constants. A second forced register graph exhibits a genuinely different layer. Retain $k = \mathbb{Q}(\sqrt{2})$, put $\eta = 3$, and supply registers

$$v^2 = \eta, \qquad u^2 = t + v. \tag{57}$$

The constant η is not a square in k. Eliminating v forces

$$Q(u, t) = (u^2 - t)^2 - \eta = u^4 - 2tu^2 + t^2 - \eta. \tag{58}$$

Theorem 6.7 (Arithmetic prime with two constant-geometric components)**.** *Let $E \subset \mathbb{P}^1_B$ be cut out by*

$$\mathcal{Q}(U, V; t) = (U^2 - tV^2)^2 - \eta V^4. \tag{59}$$

Then:

1. *$E \to B$ is an integral finite flat cover of degree four, and $Q(u, t)$ is irreducible over $k(t)$.*
2. *Over $\overline{k}(t)$, it has exactly two irreducible quadratic factors*

$$Q = (u^2 - (t + \sqrt{\eta}))(u^2 - (t - \sqrt{\eta})). \tag{60}$$

   Each factor remains irreducible over $\overline{k}(t)$. It splits into two sections only after the further nonconstant Kummer base change adjoining $\sqrt{t \pm \sqrt{\eta}}$.

3. *Its discriminant in u is*

$$\operatorname{disc}_u Q = 256\eta^2(t^2 - \eta). \tag{61}$$

   Thus the cover is finite étale away from $t^2 = \eta$. Constant Galois interchanges the two factors in (60), while analytic monodromy around $t = -\sqrt{\eta}$ or $t = +\sqrt{\eta}$ acts within the corresponding quadratic cover by $u \mapsto -u$.

4. *For h represented by M_h, put $L_1 = dZ - bW$ and $L_0 = aW - cZ$. The projective endpoint divisor of $h(u)$ is the binary quartic*

$$\mathcal{Q}_h(Z, W; t) = \left(L_1^2 - tL_0^2\right)^2 - \eta L_0^4. \tag{62}$$

   It equals $h_{B*}E$, so the same composition rule as (46) holds.

Proof. The form $\mathcal{Q}$ has no point over $V = 0$, and its affine coordinate ring is the free rank-four R-module defined by the monic polynomial Q. In the tower

$$k(t) \subset k(\sqrt{\eta})(t) \subset k(\sqrt{\eta})(t)(\sqrt{t + \sqrt{\eta}}),$$

the first extension has degree two because η is nonsquare in k, and the second has degree two because $t+\sqrt{\eta}$ has a simple zero and hence is not a square. The total degree is four, so Q is the minimal polynomial of u over $k(t)$. This also proves integrality.

After extending constants, $\sqrt{\eta}$ is available and gives (60); the simple-zero argument proves that both quadratics remain irreducible. For a polynomial u^4+pu^2+r, the discriminant is $16r(p^2-4r)^2$. Substitution of $p=-2t$ and $r=t^2-\eta$ gives (61). The Galois and local monodromy statements follow directly from the two displayed square roots. Finally, $[L_1 : L_0]=[u : 1]$ on the graph of $h(u)$; homogenized elimination gives (62). $\square$

There is also an explicit good-reduction dictionary. Work over $\mathcal{O}_k[1/6,t,t^{-1}]$ and away from $t^2=\eta$. At a closed point with residue field $\mathbb{F}_q$, let χ denote its quadratic character. If $\chi(\eta)=+1$, choose $s\in\mathbb{F}_q$ with $s^2=\eta$. Frobenius preserves the two quadratic components, and on the component $u^2=t_b\pm s$ it is the identity or a transposition according as $\chi(t_b\pm s)=+1$ or -1. If $\chi(\eta)=-1$, Frobenius exchanges the two constant components. Choosing $s\in\mathbb{F}_{q^2}$ with $s^2=\eta$, its square acts within $u^2=t_b+s$ by the character

$$\chi_{\mathbb{F}_{q^2}}(t_b+s)=\chi_{\mathbb{F}_q}(t_b^2-\eta). \tag{63}$$

Consequently Frobenius has cycle type (2)(2) when the right-hand side is $+1$, and a four-cycle when it is -1. Projective pushforward by h does not change these cycle types.

The quartic example supplies the algebraic part of the test requested in the arithmetic-versus-geometric zero-path problem: arithmetic irreducibility, constant-field components, local sheet monodromy, discriminant, and Frobenius are all explicit. It does not yet supply an AEG assignment or prove that these registers arise canonically from unrestricted expression histories.

6.6 Boundary of the restricted result

The two constructions establish an exact but narrow chain

$$\boxed{\begin{array}{c}\text{supplied algebraic register} \longrightarrow \text{marked } q=4 \text{ projective computation}\\ \downarrow\\ \text{tagged trace and pole divisor} \longrightarrow \text{projective endpoint divisor}\end{array}}$$

The first three arrows retain the declared register, time, transitions, and ordinary-domain audit. Forgetting to the final divisor retains at most the embedded projective operator and its arithmetic cover; forgetting the embedding retains only the cover. Thus this section gives a restricted Γ_4-equivariant assignment, not the general history-to-divisor functor.

Open problem 6.8 (Descent of trace data without tautological history labels). Find a category of typed algebraic registers and marked arithmetic histories in which the tagged trace and pole cocycle descend under a declared nontrivial history equivalence, while retaining enough information to distinguish ordinary domain failures and zero-hit stages. Determine whether any finite prime divisor, normalization, or stateful local system can carry this information without simply reattaching the complete word as a label. Then compare the resulting construction with the automorphic $q=4$ zero network and with parameter concatenation.

Until this descent problem is solved, the present results should be read as a proved minimal naturality model and a proved obstruction: the terminal endpoint divisor constructed here, because it factors through ρ, cannot be history-faithful, even though a computation-graph decoration can retain strictly more than its terminal operator.

7 Parameterized incidence and discriminants

7.1 Ambient families and vertical transversality

An ambient family is the correct starting object; a tube is a conclusion about a particular incidence inside it.

Definition 7.1 (Ambient AEG family)**.** An *ambient regular AEG family* over a smooth manifold B consists of a smooth fiber bundle $q : X \to B$ whose vertical tangent bundle $T^V X$ carries a smooth orientation, a smooth vertical Riemannian metric g^V, and a smooth function $A : X \to \mathbb{R}$ such that $(M_b, g_b, A_b; \mu, \lambda)$ is a regular AES for every $b \in B$. Its assignment incidence is $\mathcal{Z}(A) = A^{-1}(0)$.

Equivalently, the structure group of the surface bundle is reduced to orientation-preserving diffeomorphisms. Merely choosing an orientation on each fiber without requiring those choices to vary continuously would not orient $T^V X$ and is not sufficient here.

More generally, a *field family over an AEG background* is a rank-r real vector bundle $E \to X$ with a smooth section s. It need not be the real assignment and need not satisfy the eikonal equation. A complex-valued field is the case $r = 2$ after forgetting complex scalar multiplication.

Let $T^V X = \ker(\mathrm{d}q)$ be the vertical tangent bundle. At a zero of s, choose any connection to split TX and define the vertical derivative

$$D^V s : T^V_x X \longrightarrow E_x.$$

Its value at a zero is independent of the chosen local trivialization or connection: it is the derivative of the restricted section on the fiber.

Theorem 7.2 (Parameterized zero-section theorem)**.** *Let $q : X \to B$ be a smooth fiber bundle and $E \to X$ a rank-r vector bundle. Suppose the section $s \in \Gamma(E)$ is vertically transverse to the zero section, meaning that $D^V s$ is surjective at every $x \in \mathcal{Z}(s)$. Then:*

1. *$\mathcal{Z}(s)$ is a smooth submanifold of X of codimension r;*
2. *the restricted projection*
$$\pi = q|_{\mathcal{Z}(s)} : \mathcal{Z}(s) \longrightarrow B$$
is a smooth submersion;
3. *if π is surjective and proper, then it is a locally trivial smooth fiber bundle.*

The third conclusion has the standard relative form when X has boundary and the zero incidence is neat, with the boundary restriction also a submersion.

Proof. Vertical surjectivity implies surjectivity of the full derivative of s, so the zero-section transversality theorem gives the first conclusion. To prove the second, take $\xi \in T_b B$ and a lift $w \in T_x X$. Since $D^V s$ is surjective, there is $v \in T^V_x X$ with $D^V s(v) = -Ds(w)$. Then $w + v \in T_x \mathcal{Z}(s)$ and $Dq(w + v) = \xi$, so $D\pi$ is surjective. The third conclusion is Ehresmann's fibration theorem; the neat relative version follows by the corresponding boundary argument [9]. □

The rank is decisive. If the fibers of q have dimension m, then a regular zero fiber has dimension $m - r$.

Fiber dimension	Field rank	Fiberwise zero	Natural transport
2	1	curves	component and mapping-class monodromy
2	2	points	permutation and braid monodromy

For an ambient regular AEG family, equation (AEG) makes $D^V A = \mathrm{d}A_b$ nonzero everywhere, so the first two conclusions are automatic at zero. Properness is not.

7.2 The discriminant is not a single equation

For finite-dimensional polynomial root families, the algebraic discriminant detects repeated roots. A general AEG family has more ways to lose topological stability.

Definition 7.3 (Structural discriminant)**.** For a parameterized zero problem, the *structural discriminant* is the union of the following loci in the parameter space, whenever they are present:

$$\begin{aligned}
\mathcal{D}_{\mathrm{crit}} &= \{b : D^V s \text{ fails to be surjective at some zero over } b\},\\
\mathcal{D}_{\partial} &= \{b : \text{the zero incidence fails the specified neat boundary condition}\},\\
\mathcal{D}_{\mathrm{sing}} &= \{b : \text{a zero meets the declared singular stratum or the metric degenerates}\},\\
\mathcal{D}_{\mathrm{dom}} &= \{b : \text{the domain, rank, or regularity class of the field changes}\},\\
\mathcal{D}_{\infty} &= \{b : \pi \text{ is not proper over any neighborhood of } b\}.
\end{aligned}$$

If projective continuation is used, its ordinary poles and points at infinity are recorded separately rather than silently absorbed into $\mathcal{D}_{\mathrm{sing}}$.

This definition is a bookkeeping union, not a claim that every stratum is a smooth hypersurface. The geometry of its intersections and a canonical Whitney stratification remain open in general.

For the monic relative divisor of Theorem 4.3, the algebraic discriminant is the scheme-theoretic analogue of the collision/ramification locus, and its complement supports a finite étale root covering. If $K \subset \mathbb{C}$, then on a smooth analytic open of B^{an} where the polynomial is regarded as a complex rank-2 field, this locus is the repeated-root, hence vertical-critical, part of $\mathcal{D}_{\mathrm{crit}}$. Without that analytification and smooth-locus restriction the two categories are not identified. The algebraic discriminant does not see $\mathcal{D}_{\partial}$, $\mathcal{D}_{\mathrm{sing}}$, $\mathcal{D}_{\mathrm{dom}}$, or $\mathcal{D}_{\infty}$. Thus arithmetic factorization and geometric stability fit into the same framework without identifying the polynomial discriminant with the entire structural discriminant.

Corollary 7.4 (Stability off the discriminant)**.** *Let $U \subset B$ be connected. Over U, assume that $q : X \to B$ and the rank-r bundle E remain fixed smooth bundles, that s is vertically transverse to zero, and that the incidence is surjective and locally proper over U. If a boundary is present, also assume that the incidence is neat and that its boundary projection is a submersion. Then $\pi : \mathcal{Z}(s)|_U \to U$ is a smooth fiber bundle. In particular, the diffeomorphism type and number of connected components of the fiber are locally constant on U. These are precisely the operative hypotheses encoded by saying that U avoids the relevant pieces of the structural discriminant.*

Proof. For each $b \in U$, local properness supplies a neighborhood V on which the restricted incidence projection is proper. The other hypotheses make that restriction a surjective submersion, including the relative boundary condition when needed. Theorem 7.2 therefore gives a bundle chart over a possibly smaller neighborhood of b. Such charts are exactly the local-triviality assertion. $\square$

7.3 Escape to infinity without a critical zero

The following regular AEG family shows why $\mathcal{D}_{\infty}$ cannot be omitted. Set

$$h_t(y) = y(ty-1), \qquad A(x,y,t) = e^x h_t(y),$$

and equip each copy of $\mathbb{R}^2$ with the smoothly varying conformal metric

$$g_t = \frac{e^{2x}(h_t(y)^2 + (2ty-1)^2)}{\mu^2 + \lambda^2 e^{2x} h_t(y)^2}(\,\mathrm{d}x^2 + \,\mathrm{d}y^2). \tag{64}$$

At a zero of h_t, its derivative $2ty-1$ equals -1 or 1, so it never vanishes. More strongly, h_t and $2ty-1$ never vanish simultaneously anywhere. Thus every slice is a regular AES by Theorem 2.2.

Its zero fibers are

$$Z(A_t) = \begin{cases} \{y=0\}, & t=0, \\ \{y=0\} \amalg \{y=1/t\}, & t \neq 0. \end{cases} \tag{65}$$

The total zero incidence is the disjoint union of the plane $y=0$ and the hyperbolic sheet $ty=1$. It is smooth, and its projection to t is a submersion. Nevertheless the second component escapes to spatial infinity as $t \to 0$.

Proposition 7.5 (Nonproper topology change)**.** *The family* (64) *has no critical zero and no singular metric, but the number of zero components changes at* $t=0$*. Its incidence projection is not proper over any neighborhood of* 0*.*

Proof. Only the last statement remains. For $t_j \to 0$ with $t_j \neq 0$, the points $(0, 1/t_j, t_j)$ lie in the incidence above a compact parameter interval and have no convergent subsequence. Hence the inverse image of that interval is not compact. □

Thus topology change does not require $\mathrm{d}A_t = 0$ at a finite zero. It may occur at infinity. Conversely, a smooth total incidence does not imply a tube.

7.4 Boundary control

If a surface fiber has boundary, a zero curve may be closed or may end on the boundary. For a stable relative tube one requires the incidence to meet $\partial^V X$ neatly and the restriction $\pi : \partial\mathcal{Z} \to B$ to remain a proper submersion. Without this condition a zero component can be created by tangency to the boundary even though the interior field remains regular. Such a parameter lies in $\mathcal{D}_\partial$, not in $\mathcal{D}_{\mathrm{crit}}$.

Open problem 7.6 (Stratified AEG discriminant)*.* Under natural completeness and asymptotic hypotheses, determine when the five loci in Definition 7.3 admit a locally finite Whitney stratification and when a Thom-type isotopy lemma applies to the combined AEG data.

8 Proper zero tubes

8.1 The proper-tube theorem

Definition 1.2 reserves the word tube for a proper submersion. The resulting structure is strong and concrete.

Theorem 8.1 (Proper real-zero tube)**.** *Let $q : X \to B$ be an ambient regular AEG family over a connected one-dimensional base, and suppose its assignment incidence $\mathcal{Z} = A^{-1}(0)$ is nonempty and $\pi = q|_{\mathcal{Z}}$ is proper and surjective.*

1. *The map $\pi : \mathcal{Z} \to B$ is a smooth fiber bundle whose fibers are smooth one-manifolds.*

2. *If the surface fibers have no boundary, every zero fiber is a finite disjoint union of circles.*

3. If B is an interval, the bundle is globally trivial. For a product ambient family $M \times B$, its compact zero curves over each compact subinterval form a smooth isotopy and hence extend to an ambient isotopy of M. If B itself is a compact interval, this applies over all of B.

4. If $B = S^1$, the surface fibers are boundaryless, and the smooth vertical orientation required in Definition 7.1 is fixed, every connected component of $\mathcal{Z}$ is a torus.

Proof. The AEG equation makes the vertical derivative $\mathrm{d}A_b$ nonzero at every point, so Theorem 7.2 proves the bundle assertion. Properness makes every fiber compact. A compact boundaryless one-manifold is a finite union of circles.

A smooth bundle over an interval is trivial. In a product ambient family, a trivialization expresses the inclusions of the compact fiber as a smooth isotopy of embeddings over each compact subinterval; the isotopy-extension theorem extends it to the ambient surface [13].

Over S^1, the total space is the mapping torus of a diffeomorphism of a finite union of circles. Each orbit of the induced component permutation produces one connected component. After traversing the orbit, the return map is a diffeomorphism of S^1, so that component is either a torus or a Klein bottle. The smooth orientation of $T^V X$ and the orientation of the base orient X; the regular level set $A^{-1}(0)$ is cooriented by $\mathrm{d}A$, hence orientable. The Klein-bottle case is excluded, and every component is a torus. This is where the global vertical-orientation hypothesis is essential: a mapping torus with orientation-reversing fiber monodromy need not be orientable. □

Remark 8.2 (Why vertical orientability is necessary). If the global vertical orientation is dropped, the torus conclusion is false even for a product-type formula. Let $M = \mathbb{R}_r \times S^1_\phi$ with $g = \mathrm{d}r^2 + \mathrm{d}\phi^2$, $a = \mu r$, and $\lambda = 0$, and form the mapping torus of the AEG isometry $(r, \phi) \mapsto (r, -\phi)$. Its zero incidence is the mapping torus of an orientation-reversing circle map, hence a Klein bottle. It is compact and proper over the parameter circle, but the resulting surface bundle has no smooth orientation of $T^V X$.

Remark 8.3 (What this theorem does not encode). The theorem classifies the abstract total surface under its hypotheses. The embedding $\mathcal{Z} \hookrightarrow X$ may still carry nontrivial ambient monodromy, and no preferred curve inside a torus component has been selected. In particular, a union of zero tori is not yet a knot or link.

Example 2.1 gives an immediate nonempty instance. If B is compact and $c : B \to \mathbb{R}$ is smooth, set $a_b = a_{c(b)}$. Then

$$\mathcal{Z} = \{(r, \phi, b) : r = c(b)\} \cong S^1 \times B.$$

It is compact and therefore proper over B. The example is deliberately trivial: it verifies every hypothesis while isolating the content supplied by properness.

8.2 A compact helical AEG tube

The next family is still explicit but has nontrivial component transport. Let

$$M_L = [-L, L]_x \times S^1_y, \qquad B = S^1_\theta, \qquad L > 0,$$

and choose integers $n \geq 1$ and $m \in \mathbb{Z}$. Define

$$A_{m,n}(x, y, \theta) = e^{nx} \sin(ny - m\theta) \tag{66}$$

and the vertical metric

$$g_{m,n,\theta} = \frac{n^2 e^{2nx}}{\mu^2 + \lambda^2 e^{2nx} \sin^2(ny - m\theta)} (\mathrm{d}x^2 + \mathrm{d}y^2). \tag{67}$$

Theorem 8.4 (Helical zero-tube theorem)**.** *The family (66)–(67) is an ambient regular AEG family and its zero incidence is a proper neat tube over S^1. For every θ, the zero fiber consists of $2n$ intervals*

$$y = \frac{m\theta + k\pi}{n}, \qquad k \in \mathbb{Z}/(2n). \tag{68}$$

Parameter monodromy acts on the component labels by

$$k \longmapsto k + 2m \pmod{2n}. \tag{69}$$

If $d = \gcd(n, m)$, including $d = n$ when $m = 0$, the total incidence is a disjoint union of $2d$ annuli, each containing n/d zero intervals in its monodromy orbit.

Proof. The spatial derivative is

$$\mathrm{d}^V A_{m,n} = ne^{nx} \sin(ny - m\theta)\, \mathrm{d}x + ne^{nx} \cos(ny - m\theta)\, \mathrm{d}y,$$

whose Euclidean norm squared is $n^2 e^{2nx}$. Thus (67) is precisely the conformal realization metric, proving the AEG equation. At a zero, the y-derivative is nonzero, so the incidence is smooth and meets $x = \pm L$ neatly. It is closed in the compact manifold $M_L \times S^1$, hence compact and proper.

Equation (68) follows from $ny - m\theta \in \pi\mathbb{Z}$. Increasing θ by 2π changes the label by $2m$, proving (69). Translation by $2m$ on the cyclic group of order $2n$ has $\gcd(2n, 2m) = 2d$ orbits, each of length n/d. The transport fixes the x-coordinate, so the return map after a complete label orbit is the identity on the interval. Its mapping torus is therefore an annulus, not a Möbius band. $\square$

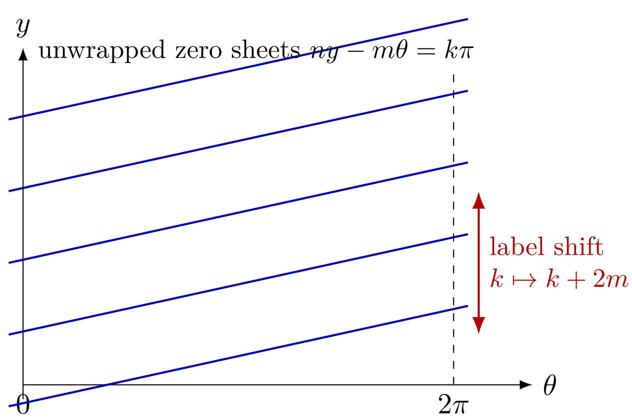

Figure 2: The helical tube on the unwrapped (θ, y)-plane. Vertical boundary identification at $\theta = 0, 2\pi$ produces the permutation (69); the drawing is schematic, while the slope is fixed exactly by $\mathrm{d}y/\mathrm{d}\theta = m/n$.

8.3 Boundary traces and logarithmic sheets

At either boundary component $x = \pm L$, the trace lies in the parameter-position torus $S^1_\theta \times S^1_y$ and satisfies

$$ny - m\theta \in \pi\mathbb{Z}.$$

It has $2d$ connected components. Each is a primitive curve with homology class

$$\frac{n}{d}[S^1_\theta] + \frac{m}{d}[S^1_y]. \tag{70}$$

If $T(p,q)$ denotes the torus link whose total homology data in these ordered coordinates are (p,q), the full trace is $T(2n,2m)$, up to simultaneous orientation reversal. This statement depends on the declared torus coordinates; it is not an intrinsic knot type before an embedding of that torus in a three-manifold is fixed.

Unwrap the y-circle to $y \in \mathbb{R}$. Formula (68) now has labels $k \in \mathbb{Z}$. The parameter loop sends $k \mapsto k+2m$, while the deck transformation $y \mapsto y+2\pi$ sends $k \mapsto k+2n$. The finite downstairs system is exactly the quotient of this countable label set by the deck action. This simultaneously realizes a proper tube and a nontrivial logarithmic shift.

Corollary 8.5 (Finite quotient of the helical lift)**.** *For the helical family, the countable lifted sheets contain no additional component permutation beyond the affine action generated by the two translations $2m$ and $2n$ on $\mathbb{Z}$. Downstairs component monodromy is their reduction modulo $2n$.*

8.4 Embedded monodromy versus braid monodromy

The permutation (69) is honest monodromy, but it is not an element of B_{2n}. Its fiber consists of intervals in a surface, not points in a disc. Intersecting the zero tube with $x=c$ produces a one-dimensional trace in $S^1_y \times S^1_\theta$, but the choice of c is additional probe data. Section 11 formalizes this distinction.

Even for compact circle fibers in Theorem 8.1, the abstract monodromy of the one-manifold fiber records only a permutation and orientation-preserving return maps of circles. The embedding in the ambient surface can carry a richer mapping class, but that is a mapping-class problem for curves, not the ordinary braid group of moving points.

9 Singular fibers and branched fillings

9.1 Real Morse events in the singular category

Let $\tau \in \mathbb{R}$ and consider the two functions

$$a^+_\tau(x,y) = x^2+y^2-\tau, \qquad a^-_\tau(x,y) = x^2-y^2-\tau. \tag{71}$$

Their common critical set in each slice is the origin. On $\mathbb{R}^2 \setminus \{0\}$ define

$$g^\pm_\tau = \frac{4(x^2+y^2)}{\mu^2+\lambda^2(a^\pm_\tau)^2}(\,\mathrm{d}x^2+\,\mathrm{d}y^2). \tag{72}$$

Proposition 9.1 (AEG Morse bifurcations)**.** *For every τ, the tuple*

$$(\mathbb{R}^2, \{0\}, g^\pm_\tau, a^\pm_\tau; \mu, \lambda)$$

is a singular AES. Its total zero incidence in $\mathbb{R}^2 \times \mathbb{R}_\tau$ is smooth, but projection to τ has a critical point at $(0,0,0)$.

For a^+_τ, the zero changes from empty to a singular point and then to a circle as τ crosses zero. For a^-_τ, the two hyperbolic branches reconnect through the four regular rays meeting at the singular origin when $\tau=0$.

Proof. The Euclidean norm squared of $\mathrm{d}a^\pm_\tau$ is $4(x^2+y^2)$, so Corollary 2.3 gives the AEG equation and local essentiality. Define $A^\pm(x,y,\tau) = a^\pm_\tau(x,y)$. Since $\partial_\tau A^\pm = -1$, zero is a regular value of the total function; hence $(A^\pm)^{-1}(0)$ is smooth. The parameter projection restricted to it is critical precisely when the spatial derivative vanishes, which happens at the origin. The stated zero sets follow directly from $x^2+y^2=\tau$ and $x^2-y^2=\tau$. $\square$

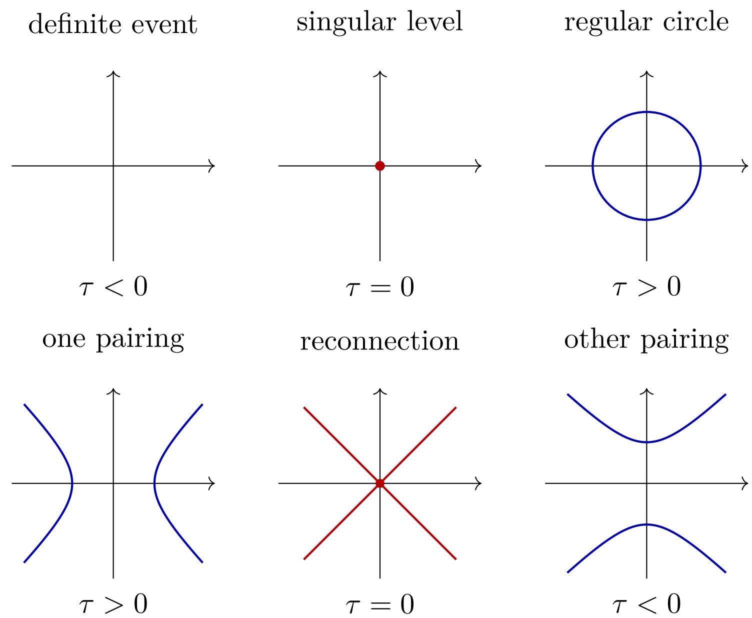

Figure 3: Real AEG-compatible Morse examples. The red point is a declared essential singularity. The pictures record zero topology only; they are not a classification of metric germs.

The proposition exhibits the key phenomenon

$$\text{smooth total zero incidence} \not\Longrightarrow \text{zero tube.}$$

At the event, it is the projection that fails to be a submersion.

9.2 A regular-to-singular fold wall

A related family makes all noncritical slices regular rather than retaining a singular point away from the zero. Put

$$A(x, y, \tau) = e^x(y^2 - \tau).$$

For $\tau \neq 0$, the spatial derivative never vanishes; for $\tau = 0$, it vanishes along the entire line $y = 0$. On the regular locus use

$$g_\tau = \frac{e^{2x}\big((y^2-\tau)^2 + 4y^2\big)}{\mu^2 + \lambda^2 e^{2x}(y^2-\tau)^2}\,(\mathrm{d}x^2 + \mathrm{d}y^2). \tag{73}$$

For $\tau < 0$ the zero fiber is empty, for $\tau > 0$ it consists of two parallel lines, and for $\tau = 0$ it is the singular line $y = 0$. The metric degenerates there and the line is locally essential. This is a verified wall-crossing family, but it is nonproper and its singular stratum is one-dimensional.

9.3 The complex branch model

Real scalar zeros and complex roots have different local models. Let $(w, \tau) \in \mathbb{C}^2$ and consider

$$F(w, \tau) = w^2 - \tau. \tag{74}$$

The incidence $F^{-1}(0)$ is a smooth complex curve parameterized by $w \mapsto (w, w^2)$. Its projection to the τ-plane is a two-sheeted branched cover with a single simple branch point at the origin.

On the boundary circle $\tau = \varepsilon e^{i\theta}$, the two roots are

$$w_\pm(\theta) = \pm\sqrt{\varepsilon}\, e^{i\theta/2}. \tag{75}$$

They remain distinct for every boundary parameter and exchange after one circuit. The boundary therefore carries the standard half-twist in B_2; the collision is at the center of the filling, not on the boundary braid.

Proposition 9.2 (Simple polynomial discriminant normal form)**.** *Let a holomorphic family of monic polynomials, parameterized by a complex d-manifold, meet the polynomial discriminant transversely at a parameter where exactly one root has multiplicity two and all other roots are simple. After holomorphic changes of root and parameter coordinates $\tau = (\tau_1, \dots, \tau_d)$, its local root incidence is the disjoint union of simple graphs and one copy of*

$$w^2 = \tau_1,$$

with $\tau_2, \dots, \tau_d$ as spectator coordinates.

Proof. Separate the simple roots by the holomorphic implicit-function theorem. The remaining quadratic factor has the form $w^2 + b(\tau)w + c(\tau)$. Translating w removes the linear term. Its discriminant is a holomorphic function of the parameter; transversality makes that function the first member τ_1 of a local coordinate system. Rescaling τ_1 gives (74), while the remaining parameter coordinates pass through unchanged. □

Restricting τ to the real axis gives two real roots for $\tau > 0$ and two imaginary roots for $\tau < 0$. Their four half-arcs meet at the branch point. This four-valent picture is a real slice of a complex branched cover, not a regular four-valent zero of a real assignment.

There is nevertheless a precise relation to Theorem 2.4. If the same local map is regarded spatially as $\Phi(z) = z^2$ and one takes the real assignment $\operatorname{Im}\Phi$, its zero germ is four-pronged and its pulled-back AEG metric has cone angle 4π. In the present subsection, by contrast, $w^2 = \tau$ is an incidence over a parameter plane and the output is root monodromy. The same holomorphic branch supplies two constructions only after the roles of source, target, and parameter have been declared.

9.4 Smooth and holomorphic fillings

Section 10 realizes every braid by a family that is smooth in the loop parameter and holomorphic in the spatial coordinate w. Extending such a family over a parameter disc while remaining holomorphic in both variables is a stronger demand. Algebraic-function boundaries are constrained by quasipositivity [17]. Thus the statements

every braid has an arithmetic-holomorphic spatial realization,
every braid bounds a holomorphic AEG root filling

are not equivalent; the first is proved below, while the second is false without restrictions.

Open problem 9.3 (AEG-natural singular fillings)*.* Determine which branch signs and global braided surfaces arise when the parameter dependence is constrained by expression history or by a natural connection, rather than chosen as an arbitrary smooth or holomorphic polynomial family.

10 Complex roots, monodromy, and braids

10.1 Square-free polynomials as a finite-root interface

Let $\mathcal{P}_n \cong \mathbb{C}^n$ be the space of monic degree-n complex polynomials and let $\Delta_n \subset \mathcal{P}_n$ be its algebraic discriminant. The square-free locus is

$$\mathcal{P}_n^{\mathrm{sf}} = \mathcal{P}_n \setminus \Delta_n.$$

Vieta's map identifies it with the unordered configuration space

$$\mathcal{P}_n^{\mathrm{sf}} \cong \mathrm{UConf}_n(\mathbb{C}) = \mathrm{Conf}_n(\mathbb{C})/S_n. \tag{76}$$

Its fundamental group is the Artin braid group B_n [2, 10, 4].

This is the complex-analytic specialization of the relative zero-divisor picture in Theorem 4.3. Irreducible factors over the coefficient field record arithmetic components; after passing to a complex parameter manifold and removing the discriminant, the geometric sheets form the covering used below. Neither passage identifies sheets with raw expression words.

Theorem 10.1 (Finite-root incidence and braid monodromy)**.** *Let B be a connected smooth manifold and let*

$$F_b(w) = w^n + c_1(b)w^{n-1} + \cdots + c_n(b) \tag{77}$$

be a smooth family of monic polynomials with nonvanishing discriminant. Then

$$\mathcal{R}_F = \{(w, b) \in \mathbb{C} \times B : F_b(w) = 0\} \tag{78}$$

is a smooth real codimension-two submanifold whose projection to B is a proper n-sheeted covering. The coefficient map $c : B \to \mathcal{P}_n^{\mathrm{sf}}$ induces $c_* : \pi_1(B, b_0) \to \pi_1(\mathcal{P}_n^{\mathrm{sf}}, c(b_0))$. After choosing the usual path to a standard base configuration and the resulting identification of the target with B_n, this is braid monodromy

$$c_* : \pi_1(B, b_0) \longrightarrow B_n. \tag{79}$$

Changing that identification conjugates the representation. Its composition with $B_n \to S_n$ is the ordinary permutation monodromy of the covering $\mathcal{R}_F \to B$.

Proof. At a root of a square-free polynomial, $\partial F_b/\partial w \neq 0$. The implicit-function theorem therefore writes the root incidence locally as n disjoint graphs over B, so it is an n-sheeted covering. A finite covering of locally compact Hausdorff spaces is proper. The map to unordered root configurations is exactly the coefficient map under (76); functoriality of the fundamental group gives (79). Forgetting paths and retaining only their endpoint labels is the standard quotient $B_n \to S_n$. □

For $B = S^1$, the incidence is a closed geometric braid in $\mathbb{C} \times S^1$. A basepoint and an ordering above it give an element of B_n; changing the basepoint or ordering conjugates that element. This is strictly more information than the endpoint permutation.

Warning 10.2 (Rank-one assignments do not yield B_n)*.* The field F in Theorem 10.1 is complex-valued and has real rank two. The theorem does not turn a real AEG assignment into a polynomial, nor does it promote a moving family of zero curves to a finite-root braid. The two constructions share the parameterized zero-section theorem but have different codimensions.

10.2 Arithmetic-holomorphic braid realization

On the complete basic hyperbolic AES with $\mu, \lambda > 0$, Paper II constructs a global coordinate $w = X + iy \in \mathbb{H}$ in which arithmetic holomorphicity is ordinary holomorphicity. This supplies a precise AEG background for Theorem 10.1.

Theorem 10.3 (AEG realization of every braid)**.** *For every $\beta \in B_n$ and every complete basic hyperbolic AES with $\mu, \lambda > 0$, there is a smooth loop of arithmetic-holomorphic fields F_θ such that:*

1. *every F_θ is a monic degree-n polynomial in the arithmetic holomorphic coordinate w;*
2. *all roots lie in a fixed disc $D \Subset \mathbb{H}$ and remain simple;*
3. *the root incidence realizes the geometric braid β.*

Consequently, every oriented link is the closure of the root incidence of such an arithmetic-holomorphic family.

Proof. Represent β by a smooth loop $\{z_1(\theta), \dots, z_n(\theta)\}$ in $\mathrm{UConf}_n(\mathbb{C})$. On the parameter interval $[0, 2\pi]$, choose a smooth ordered lift $(z_1(\theta), \dots, z_n(\theta))$; its two endpoints may differ by the permutation underlying β. An affine translation and positive rescaling place the compact unordered image in a disc $D \Subset \mathbb{H}$ without changing its braid class. Set

$$F_\theta(w) = \prod_{j=1}^{n} (w - z_j(\theta)). \tag{80}$$

Because the endpoint lists differ only by a permutation, the coefficients are smooth and periodic in θ. The roots are simple, and each slice is holomorphic in w. Paper II identifies it as an arithmetic-holomorphic field. Its root incidence is the chosen loop of configurations. The last statement follows from Alexander's closed-braid theorem [1]. □

The theorem is a universality result, not an invariant. Because every braid can be realized, the predicate "admits an AEG root-tube realization" cannot distinguish links. New information would have to be supplied by a natural history-derived decoration or by a restricted class of allowed families.

10.3 Logarithmic lifting of root tubes

Suppose all roots avoid the origin. Pull the root incidence back along $\exp : \mathbb{C} \to \mathbb{C}^\times$:

$$\widetilde{\mathcal{R}}_F = \{(b, \omega) : F_b(e^\omega) = 0\}. \tag{81}$$

Theorem 10.4 (Logarithmic root transport and gauge)**.** *The map $\widetilde{\mathcal{R}}_F \to \mathcal{R}_F$ is the canonical principal $2\pi i\mathbb{Z}$-cover obtained by pullback of the exponential cover. Its specified deck-translation action is $\omega \mapsto \omega + 2\pi ik$. Along a parameter loop, choose a labeling of the initial roots and logarithmic lifts. If root continuation has permutation σ, then there are integers k_i such that*

$$\omega_i(1) = \omega_{\sigma(i)}(0) + 2\pi i k_i. \tag{82}$$

Changing the initial lifts by $\omega_i(0) \mapsto \omega_i(0) + 2\pi i m_i$ changes the integers by

$$k_i \longmapsto k_i + m_i - m_{\sigma(i)}. \tag{83}$$

For every cycle of σ, the sum of the k_i on that cycle is gauge invariant. Relabeling the roots permutes these cycle sums, so their multiset, rather than an ordered list, is label-independent. The total sum $\sum_i k_i$ is gauge and label invariant and equals the winding of the product of the roots around zero.

Proof. Equation (81) is the pullback of the exponential cover, so the first claim is immediate. Unique path lifting gives (82). If the initial lift of the ith path is shifted by $2\pi i m_i$, its entire lift is shifted by that amount. Comparing the new endpoint with the new lift assigned to the root $\sigma(i)$ gives (83). The m_i terms telescope around each permutation cycle. Finally, the product of all roots is, up to sign, the constant coefficient of F_b; summing logarithmic increments computes its winding. □

Thus a vector (k_i) is not an invariant before a lift gauge is fixed. Cycle sums are invariant but abelian. The full noncommutative information remains in the configuration path, more precisely in the annular braid group when roots are constrained to $\mathbb{C}^\times$.

10.4 Boundary braid and discriminant filling

Theorem 10.1 applies only in the complement of Δ_n. A filling disc may meet the discriminant. At a transverse simple intersection, Proposition 9.2 gives $w^2 = \tau$, and its boundary loop contributes a conjugate of a standard positive or negative half-twist according to orientation. The collision is therefore a two-cell event in the filling, while the boundary braid stays collision-free.

This separation resolves an otherwise persistent ambiguity:

Object	Allowed local event	Output
regular real assignment family	no critical zero	curve tube
singular real family	Morse critical zero	birth or reconnection
square-free boundary root family	no repeated root	braid monodromy
root filling through Δ_n	simple double root	branch half-twist

Open problem 10.5 (Natural braid monodromy)*.* Find a functorial procedure that assigns a finite complex-root field to an expression history or to a real zero tube, and prove that the resulting braid is independent of auxiliary coordinates and probes. The arbitrary polynomial realization of Theorem 10.3 does not solve this problem. In particular, a solution should be compatible with the history-to-divisor functor requested in Problem 4.6.

11 Threading, affine torsion, and the knot frontier

11.1 Threads are additional data

The real zero tube has curve fibers. To obtain moving points one must select them.

Definition 11.1 (Finite zero thread)**.** Let $\pi : \mathcal{Z} \to B$ be an embedded real zero tube over a connected one-manifold. A *finite zero thread* is a properly embedded one-dimensional submanifold $K \subset \mathcal{Z}$, with $\partial K = K \cap \pi^{-1}(\partial B)$ when the base has boundary, such that $\pi|_K : K \to B$ is a finite-sheeted covering. For $B = S^1$, this forces K to be a compact one-manifold without boundary. The pair $(\mathcal{Z}, K)$ is a *threaded zero tube.* A thread is *intrinsic* only if its construction from the stated AEG data is functorial under the declared notion of AEG isomorphism.

In the helical family, every slice $x = c$ gives

$$K_c = \mathcal{Z}(A_{m,n}) \cap \{x = c\},$$

a degree-$2n$ finite thread. Its homology and component structure are computed in (70). The choice of c is a probe choice; if one uses $c = \pm L$, the boundary is part of the domain data but the choice of boundary component remains. Selecting only one of the $2d$ components is another choice.

A rank-two root incidence such as $\mathcal{R}_F$ is already one-dimensional, so it plays the role of its own finite thread. If its moving points lie in a disc, it is an ordinary geometric braid. If they lie in an annulus or general surface, the relevant surface braid group and equivalence relation must be named.

11.2 A scalar Markov obstruction

The first novelty filter is independent of AEG.

Proposition 11.2 (Stateless additive collapse)**.** *Let A be an abelian group and $n \geq 2$. Every homomorphism $\Theta_n : B_n \to A$ factors through the exponent sum (writhe) $w : B_n \to \mathbb{Z}$:*

$$\Theta_n(\beta) = w(\beta)c_n$$

for a fixed $c_n \in A$. If the family Θ_n is compatible with the standard inclusions $B_n \hookrightarrow B_{n+1}$ and is unchanged under both Markov stabilizations $\beta \mapsto \beta\sigma_n^{\pm 1}$, then every Θ_n is zero.

Proof. In an abelian target, the braid relation $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$ implies $\Theta_n(\sigma_i) = \Theta_n(\sigma_{i+1})$. All standard generators therefore have the same image c_n, and the homomorphism is exponent sum times c_n. Inclusion compatibility identifies c_n with c_{n+1}. Positive stabilization then gives $c_{n+1} = 0$; negative stabilization gives the same conclusion. Hence the family vanishes. □

Allowing a normalization depending on strand number does not evade both signs. If $I_n(\beta) = c\,w(\beta) + dn$ in an additive coefficient group, positive and negative stabilization require $c + d = 0$ and $-c+d = 0$. Away from two-torsion this forces $c = d = 0$, and in all cases it leaves no noncommutative information. A useful AEG invariant must retain state, use a trace-like normalization, or belong to a more structured category.

11.3 Affine conjugation and the Alexander layer

Let R be a commutative ring. Write an affine transformation as (p, u), with $p \in R^\times$ and $u \in R$, acting by $x \mapsto px + u$. Thus all multipliers p, q, r below are units and all translation coordinates u, v, w lie in R. We use

$$(p, u)(q, v) = (pq, u + pv), \qquad (p, u)^{-1} = (p^{-1}, -p^{-1}u). \tag{84}$$

Right conjugation $x \triangleright y = y^{-1}xy$ is

$$(p, u) \triangleright (q, v) = \Big(p, q^{-1}(u + (p - 1)v)\Big). \tag{85}$$

On the fixed-multiplier branch $X_t = \{(t, u) : u \in R\}$ over a commutative ring in which t is a unit, this becomes

$$u \triangleright v = t^{-1}u + (1 - t^{-1})v. \tag{86}$$

It is exactly an Alexander quandle, not a structure beyond the Alexander layer.

The affine torsion expression

$$\kappa\big((p, u), (q, v)\big) = (1 - q)u + (p - 1)v \tag{87}$$

restricts to

$$\kappa_t(u, v) = (t - 1)(v - u). \tag{88}$$

Proposition 11.3 (Ordinary exactness)**.** *With the ordinary quandle coboundary convention $df(x,y) = f(x \triangleright y) - f(x)$, one has*

$$\kappa_t = d(tf) \qquad \textit{for } f(u) = u.$$

Thus the fixed-multiplier affine torsion represents the zero ordinary quandle cohomology class.

Proof. Equation (86) gives

$$df(u,v) = (t^{-1} - 1)u + (1 - t^{-1})v = \frac{t-1}{t}(v-u).$$

Multiplication by t gives (88). ☐

11.4 Twisting, resonance, and a nonzero class

We now use the twisted convention of Carter–Elhamdadi–Saito [5]. Let the coefficient action be a scalar T_{coef}. A normalized twisted 2-cocycle ϕ satisfies

$$\begin{aligned} T_{\mathrm{coef}}\phi(x,y) + \phi(x \triangleright y, z) = T_{\mathrm{coef}}\phi(x,z) + (1 - T_{\mathrm{coef}})\phi(y,z) \\ + \phi(x \triangleright z, y \triangleright z), \end{aligned} \tag{89}$$

and $\phi(x,x) = 0$. For coboundaries we choose the sign

$$\delta_T^+ f(x,y) = T_{\mathrm{coef}} f(x) + (1 - T_{\mathrm{coef}}) f(y) - f(x \triangleright y). \tag{90}$$

The chain-level convention in the cited paper differs from (90) by a global minus sign in one displayed definition, while its extension and state-sum calculations use the sign above. The coboundary subgroup is the same under either choice.

Put $\alpha = t^{-1}$. Direct substitution in (89) shows that κ_t is a twisted cocycle for every coefficient action. If $\alpha - T_{\mathrm{coef}}$ is a unit in the coefficient ring, it is exact: for

$$f(u) = \frac{t-1}{\alpha - T_{\mathrm{coef}}} u$$

one has $\delta_T^+ f = \kappa_t$. Over a field, only the resonance $T_{\mathrm{coef}} = \alpha$ survives this linear exactness test; over a general ring, a nonzero nonunit difference may also obstruct this particular linear primitive.

Theorem 11.4 (Finite-field resonant affine class)**.** *Let q be a prime power, let $t \in \mathbb{F}_q^\times \setminus \{1\}$, and put $\alpha = t^{-1}$. Give $X_t = \mathbb{F}_q$ the Alexander operation $x \triangleright y = \alpha x + (1-\alpha)y$ and give the coefficient module $N = \mathbb{F}_q$ the resonant action $T_{\mathrm{coef}} = \alpha$. Then*

$$\kappa_t(x,y) = (t-1)(y-x) \tag{91}$$

is a normalized twisted 2-cocycle representing a nonzero class in $H^2_{TQ}(X_t; N)$.

Proof. Normalization is immediate, and substituting the Alexander operation into (89) makes both sides equal

$$(t-1)\big[z - (\alpha + T_{\mathrm{coef}})x + (\alpha + T_{\mathrm{coef}} - 1)y\big].$$

Suppose $\kappa_t = \delta_T^+ f$. Subtracting the constant $f(0)$ does not change the coboundary, so take $g(0) = 0$. Setting $y = 0$ gives

$$g(\alpha x) - \alpha g(x) = (t-1)x. \tag{92}$$

Let $r = \mathrm{ord}(\alpha)$. Iterating (92) yields

$$g(\alpha^r x) = \alpha^r g(x) + r(t-1)\alpha^{r-1}x.$$

But $\alpha^r = 1$ and r divides $q-1$, so the characteristic of $\mathbb{F}_q$ does not divide r. Taking $x \neq 0$ contradicts $t \neq 1$. Hence no one-cochain has coboundary κ_t. □

This closes the algebraic resonance question, but it does not yet produce a stronger classical knot invariant. In fact, a second calculation gives a sharp collapse theorem.

We now fix the precise planar state-sum convention. Give each oriented arc the normal for which the ordered pair (tangent, normal) agrees with the plane orientation, and give the unbounded region Alexander number zero. At a crossing τ, let $L(\tau)$ be the Alexander number of the source region from which both arc normals point. Let y_τ be the over-arc color, and let its normal point from an under-arc colored x_τ to one colored $x_\tau \triangleright y_\tau$. Define $\epsilon(\tau) = 1$ when the ordered pair (over-arc normal, under-arc normal) agrees with the plane orientation and $\epsilon(\tau) = -1$ otherwise. In additive coefficient notation the twisted Boltzmann weight is

$$B_T(\tau, C) = T_{\mathrm{coef}}^{-L(\tau)}(\epsilon(\tau)\kappa_t(x_\tau, y_\tau)) \in N, \tag{93}$$

and the state sum is

$$\Phi_{\kappa_t}(D) = \sum_{C \in \mathrm{Col}_{X_t}(D)} \left[\sum_\tau B_T(\tau, C)\right] \in \mathbb{Z}[N]. \tag{94}$$

Here $[a]$ denotes the basis element of the integral group ring indexed by $a \in N$. This is the additive form of the standard twisted convention.

Theorem 11.5 (Planar state-sum collapse)**.** *Under the hypotheses of Theorem 11.4, the standard twisted quandle cocycle state sum (94) is a Reidemeister invariant of oriented classical links, but for every planar link diagram D it is*

$$\Phi_{\kappa_t}(D) = |\mathrm{Col}_{X_t}(D)|[0]. \tag{95}$$

Thus it provides no enhancement of the Alexander quandle coloring count.

Proof. The Reidemeister invariance follows from the standard twisted-cocycle theorem [5]. To identify the value, define the two-dimensional $\mathbb{F}_q$-vector space $E = N \oplus X_t$ with Laurent action

$$T_E(a, x) = (\alpha a - (t-1)x, \alpha x). \tag{96}$$

This is invertible and makes $0 \to N \to E \to X_t \to 0$ a short exact sequence of Alexander modules. For the section $s(x) = (0, x)$,

$$T_E s(x) + (1 - T_E)s(y) - s(x \triangleright y) = (\kappa_t(x, y), 0). \tag{97}$$

Hence κ_t is the obstruction cocycle of an Alexander-module extension. The planar obstruction-cocycle cancellation theorem of Carter–Elhamdadi–Saito then makes every coloring contribute the identity weight. With convention (94), the state sum is therefore exactly $|\,\mathrm{Col}_{X_t}(D)|[0]$. □

Theorems 11.4 and 11.5 illustrate why cohomological nontriviality and knot-theoretic detection must be tested separately. A nonzero class can have a trivial planar enhancement because its weights cancel through an extension. On diagrams in nonplanar surfaces, the same extension argument need not cancel; that is a different, surface-link question and is not promoted here to an ordinary S^3 invariant.

11.5 Variable multipliers and the Reidemeister-III anomaly

For the full affine quandle, the torsion (87) is generally not an ordinary 2-cocycle. With $x = (p, u)$, $y = (q, v)$, and $z = (r, w)$, define its type-III defect by

$$\mathfrak{A}_\kappa(x, y, z) = \kappa(x, y) + \kappa(x \triangleright y, z) - \kappa(x, z) - \kappa(x \triangleright z, y \triangleright z).$$

A direct calculation gives

$$\mathfrak{A}_\kappa(x, y, z) = \frac{(q - r)(r - 1)}{qr}\, \kappa(x, y). \tag{98}$$

It vanishes on a fixed-multiplier branch, as it must. On a one-component knot, conjugate meridians also have equal multipliers, so arbitrary multiplier variation along arcs is not legitimate coloring data. Nonzero use of (98) would require additional region state, a multi-component setting, a dynamical quandle, or another declared structure.

Warning 11.6 (Anomaly is not associator). A failure of the ordinary type-III cocycle equation is not automatically a higher associator or a surface-holonomy invariant. Such an interpretation requires separate pentagon/hexagon or equivalent coherence and filling-independence proofs.

11.6 A historical free-word calculation

One useful repository computation can be stated without overclaiming it. Use the historical figure-eight knot-group presentation

$$G_{4_1} = \langle a, b \mid abbbaBAAB = 1\rangle, \qquad A = a^{-1}, \quad B = b^{-1}.$$

Under the convention that $a : x \mapsto tx$, $b : x \mapsto x + 1$, uppercase letters denote inverses, and the displayed transformations compose in the fixed word order, the free-group word

$$abbbaBAAB$$

has unit multiplier and translation coordinate

$$-\Delta_{4_1}(t) = -(t^2 - 3t + 1). \tag{99}$$

This is a verified affine word calculation. The displayed generator assignment descends from the free group to G_{4_1} only on the parameter locus $\Delta_{4_1}(t) = 0$. At generic t it is not a knot invariant; it is the failure of the chosen generator assignment to satisfy the relator.

11.7 The remaining invariant gate

Theorem 10.3 supplies every braid, and Definition 11.1 supplies explicit threads in selected real tubes. Neither result selects a choice-free numerical or algebraic decoration. For a family $I_n : B_n \to \mathcal{A}$ to descend to oriented links via Markov's theorem, it must at minimum satisfy

$$\begin{aligned} I_n(\gamma\beta\gamma^{-1}) &= I_n(\beta), \\ I_{n+1}(\iota_n\beta\, \sigma_n) &= I_n(\beta), \\ I_{n+1}(\iota_n\beta\, \sigma_n^{-1}) &= I_n(\beta). \end{aligned} \tag{100}$$

after any declared normalization [4]. An AEG construction must additionally prove independence of logarithmic lift, probe, thread, spine, axis, basepoint, and expression presentation whenever those are auxiliary choices.

Open problem 11.7 (AEG Markov descent)*.* Construct a nonabelian or stateful decoration natural under AEG isomorphisms, prove the braid relations and all choice-independence statements, and find a Markov normalization satisfying (100).

Open problem 11.8 (Separation beyond Alexander and Burau)*.* After Problem 11.7 is solved, exhibit two oriented links with the same declared Alexander/Burau baseline but different AEG invariant. A stronger braid representation without closure descent does not meet this criterion.

11.8 Conclusion

The singular-zero programme now has a linked analytic, arithmetic, and topological core. Holomorphic pullback turns degree-m ramification into a $2m$-pronged essential zero and a $2\pi m$ cone. For the projective operators generated by $z \mapsto z + \sqrt{2}$ and $z \mapsto -1/z$, the normalized Hecke Hauptmodul makes this mechanism global: its $\{\infty, 4\}$ dessin is an exact singular AEG zero network, and its assignment descends with a nontrivial sign character. This replaces a postulated apeirogonal drawing by a zero graph derived from an arithmetic operator group.

The same sign character has a precise but layered knot-theoretic meaning. Its coarse completed AEG carrier is the flat cylinder, whereas the unit tangent bundle of the hyperbolic sign cover is, under the declared cusp compactification, the complement of $T(2, 4)$ in S^3. Primitive hyperbolic operator classes give classical Hecke periodic-orbit knots, with the zero dessin serving as their coding spine only after the required directed thickening. This identifies an exact arithmetic–geometry–knot interface; it does not turn the local four-prong germ into a link or produce a new knot invariant.

The relative-divisor theorem separates irreducibility over the coefficient field, geometric sheets after scalar extension, and analytic monodromy away from the discriminant. Supplied quadratic and quartic registers now make that hierarchy finite and computable: they give exact endpoint divisors, collapse kernels, discriminants, monodromy, and Frobenius cycles, while a tagged trace retains some intermediate domain data. Their terminal divisor still factors through projective evaluation. Thus they are a restricted naturality test, not the general functor from marked histories to prime relative divisors.

Downstream, regular real assignments produce curve tubes under properness; their compact circle-parameter total spaces are tori. The helical family makes component transport, deck lifting, and selected threads explicit. Singular AEG metrics realize the two elementary real Morse events, while complex root fillings isolate the branch half-twist. Paper II's function theory realizes every braid by arithmetic-holomorphic roots.

The same analysis locates the frontier. Flexible metrics and the $\sin(1/z)$ control make analytic existence abundant, while universal braid realization makes mere representability nondiscriminating. Stateless scalars collapse to writhe. Fixed affine multipliers reduce to an Alexander quandle; their ordinary torsion is exact. Resonant twisting gives a genuinely nonzero cohomology class, but its planar classical-link state sum still collapses to the coloring count. The remaining original problem is therefore not to draw another tube or to relabel a classical periodic orbit. It is to derive canonical register or divisor data from marked histories, determine what survives inside the operator-to-knot fibers, and prove that any resulting stateful decoration survives every quotient required to become a knot invariant.

A Metric, regularity, and properness calculations

A.1 Conformal scaling identity

If $g = \rho h$ on a surface, then $g^{-1} = \rho^{-1}h^{-1}$. Therefore, for any smooth real function a,

$$|\nabla a|_g^2 = |\,\mathrm{d}a|_{g^{-1}}^2 = \rho^{-1}|\,\mathrm{d}a|_h^2. \tag{101}$$

Taking $\rho = |\,\mathrm{d}a|_h^2/(\mu^2 + \lambda^2 a^2)$ proves Theorem 2.2. The construction is valid exactly where $\mathrm{d}a \neq 0$; at a critical point the displayed coefficient vanishes and is not a Riemannian metric.

For $a_h = e^x h(y)$,

$$\partial_x a_h = e^x h, \qquad \partial_y a_h = e^x h',$$

which gives Equation (6). For the helical assignment, writing $\xi = ny - m\theta$ gives

$$\partial_x A = ne^{nx}\sin\xi, \qquad \partial_y A = ne^{nx}\cos\xi,$$

so $|\,\mathrm{d}^V A|^2 = n^2 e^{2nx}$ independently of ξ. This cancellation is the reason the helical metric has the particularly simple numerator in (67).

A.2 Vertical transversality in coordinates

Locally trivialize $q : X \to B$ and $E \to X$ near a zero so that $X \cong U \times V$ and $E \cong (U \times V) \times \mathbb{R}^r$, with $U \subset \mathbb{R}^m$ a fiber chart and $V \subset \mathbb{R}^k$ a base chart. The section becomes a map $s(u, b) \in \mathbb{R}^r$. Vertical transversality says that the $r \times m$ matrix $D_u s$ has rank r. The full Jacobian $(D_u s, D_b s)$ then has rank r, proving that $s^{-1}(0)$ has dimension $m + k - r$.

To see the projection submersion explicitly, solve

$$D_u s\,\dot{u} = -D_b s\,\dot{b}$$

for $\dot{u}$ using surjectivity of $D_u s$. Then $(\dot{u}, \dot{b})$ is tangent to the incidence and projects to the arbitrary vector $\dot{b}$. This is the local linear-algebra content of Theorem 7.2.

A.3 Properness checks

For the helical model, $M_L \times S^1$ is compact and the zero incidence is the inverse image of the closed set $\{0\}$ under a continuous function. It is therefore compact, and every map from it to the Hausdorff circle is proper. At $x = \pm L$, the vertical normal to the boundary is ∂_x, while $\partial_y A \neq 0$ on the zero set. Hence the zero incidence is neat and its boundary projects submersively.

For the escape family of Proposition 7.5, fix any $\varepsilon > 0$ and set $t_j = \varepsilon/(j + 1)$. The sequence

$$(x_j, y_j, t_j) = (0, t_j^{-1}, t_j)$$

lies in the zero incidence above $[0, \varepsilon]$ and leaves every compact subset of $\mathbb{R}^2 \times [0, \varepsilon]$. This is a direct certificate that the incidence projection is not proper.

A.4 Audit records for the explicit models

The following records make all data and limitations explicit.

Cylindrical product model. Domain $\mathbb{R} \times S^1$; metric $\mathrm{d}r^2 + \mathrm{d}\phi^2$; assignment $(\mu/\lambda)\sinh(\lambda(r - c))$ for $\lambda \neq 0$ and $\mu(r - c)$ for $\lambda = 0$; singular set empty; equation verified by differentiation; zero topology one circle; compact parameter base gives a proper product incidence; status: proved.

Parallel multi-zero model. Domain $\mathbb{R}^2$; metric (6); assignment $e^x h(y)$; singular set empty when all zeros of h are simple; equation verified by conformal realization; zero topology one line for each zero of h; no parameter required; completeness not asserted; status: proved, newly constructed in this paper.

Logarithmic model. Downstairs domain $\mathbb{C}^\times$, upstairs domain $\mathbb{C}$; metrics and assignments in (8) and (9); singular set empty; equation verified by pullback; two zero components downstairs and countably many upstairs; deck action $k \mapsto k+2$; missing origin removable; status: proved covering model, not an essential singular AES.

Holomorphic branched pullback. Domain a Riemann surface Σ; map a nonconstant holomorphic $\Phi : \Sigma \to \mathbb{C}$; assignment $\operatorname{Im} \Phi$; metric (2); singular set $\operatorname{Crit}(\Phi)$; equation verified by pullback from the linear target AES; a degree-m critical zero has $2m$ prongs and a $2\pi m$ metric cone; status: proved local theorem, with no claim that an arbitrary Φ is history-natural.

Finite Blaschke bridge. Domain $\mathbb{D}$; map a finite Blaschke product B; finite divisor $B^{-1}(0)$; phase assignments and metrics as in Example 2.6; singular set $\operatorname{Crit}(B)$ for the phase model and $B^{-1}(0) \cup \operatorname{Crit}(B)$ for the pulled-back radial disc model; zero topology a finite holomorphic divisor or the preimage of a real diameter, depending on field rank; status: proved finite bridge, but its freely selected roots are not a history-derived invariant.

Essential sine control. Domain a disc with declared singular set $\{0\} \cup \{(\pi/2 + k\pi)^{-1}\}$; assignment $\operatorname{Im} \sin(1/z)$ and metric in Example 2.7; zero topology one real-axis branch and countably many circles, with four-pronged 4π cone nodes accumulating at the essential singularity; status: proved analytic-capacity example, not the arithmetic model.

$q = 4$ **Hecke–Hauptmodul model.** Domain $\mathbb{H}$; normalized Hauptmodul β with orbifold signature $(2, 4, \infty)$, finite ramification indices 2 and 4, and a simple pole at the cusp; singular set $\mathcal{E}_4 = \beta^{-1}(1)$; assignment and metric (18); equation verified in the local coordinate $\log W$; zero topology $\beta^{-1}([0, 1])$, which becomes the $\{\infty, 4\}$ Hecke graph after bivalent vertices are suppressed; order-4 nodes have four prongs and cone angle 4π upstairs; the metric descends while the assignment descends as a sign-line-bundle section; regular locus incomplete at the cone nodes, completed full quotient a complete length space with its cusp at infinite distance; status: proved from the normalized automorphic quotient, at the projective-operator history level only.

Helical model. Domain $[-L, L] \times S^1$; parameter $\theta \in S^1$; metric (67); assignment (66); singular set empty; equation verified above; each zero fiber has $2n$ intervals; total incidence has $2 \gcd(n, m)$ annuli and is proper with neat boundary; status: proved.

Morse models. Domain $\mathbb{R}^2$, singular set $\{0\}$, parameter $\tau \in \mathbb{R}$; metric (72); assignments (71); equation verified on the punctured plane; definite birth/death and indefinite reconnection; the total zero incidence is smooth but its parameter projection is critical; status: proved examples, not universal AEG normal forms.

Complex branch model. Background is the basic hyperbolic AES from Paper II with $\mu, \lambda > 0$; field $F_\tau(w) = w^2 - \tau$ is complex-valued and arithmetic holomorphic in w; the field is not the real assignment; root incidence is smooth, projection has one simple branch point; boundary gives a half-twist; status: proved local root model.

A.5 Completeness warning

Conformal realization does not preserve completeness. If a conformal coefficient ρ decays rapidly along an end, an h-infinite path may have finite $g = \rho h$ length. In the parallel examples, the factor e^{2x} commonly makes $x \to -\infty$ a finite-distance end. Any theorem that needs complete fibers must therefore add completeness as a hypothesis and recheck each model; it cannot infer it from the eikonal identity.

The Hecke model illustrates the required distinction. Its regular locus is incomplete because the deleted order-4 points lie at finite distance. The singular length completion adjoins 4π cone points, while the cusp coordinate has metric asymptotic to $|\,\mathrm{d}q/q|^2$ and is at infinite distance. This audited completion does not turn a cone point into a smooth regular AES point.

B Configuration spaces, logarithmic lifts, and branch signs

B.1 Configuration and braid conventions

The ordered configuration space of n points in a surface S is

$$\mathrm{Conf}_n(S) = \{(z_1, \dots, z_n) \in S^n : z_i \neq z_j \text{ for } i \neq j\},$$

and $\mathrm{UConf}_n(S) = \mathrm{Conf}_n(S)/S_n$. For the plane, $\pi_1(\mathrm{UConf}_n(\mathbb{C})) = B_n$. Our generators σ_i exchange the ith and $(i+1)$st points by a positive half-twist and satisfy

$$\begin{aligned} \sigma_i\sigma_j &= \sigma_j\sigma_i \qquad\qquad (|i-j| \geq 2), \\ \sigma_i\sigma_{i+1}\sigma_i &= \sigma_{i+1}\sigma_i\sigma_{i+1}. \end{aligned}$$

The quotient $B_n \to S_n$ remembers only endpoint permutation, and the abelianization $B_n \to \mathbb{Z}$ sends every σ_i to 1.

The identification (76) sends an unordered set of roots to the elementary symmetric coefficients of its monic polynomial. It is a homeomorphism, with smooth complex-manifold structure on the square-free locus. The classical configuration-space fibrations supply one standard route to the fundamental-group computation [10].

B.2 Properness of a finite root covering

Let $p : Y \to B$ be a finite-sheeted covering with B locally compact Hausdorff. For a compact set $K \subset B$, cover K by finitely many relatively compact evenly covered neighborhoods. The closure of each sufficiently small sheet is homeomorphic to a compact closure downstairs. A finite union contains $p^{-1}(K)$ as a closed subset, so $p^{-1}(K)$ is compact. Thus p is proper. Applied to the root incidence, this proof uses square-freeness locally and the fixed finite degree globally. Monicity prevents a root from disappearing to infinity while the coefficient remains at a fixed point of B.

B.3 Gauge calculation for logarithmic roots

Let $z_i(s)$ be the continuation path beginning at the ith labeled root. If $z_i(1) = z_{\sigma(i)}(0)$ and ω_i is its logarithmic lift, then

$$\omega_i(1) - \omega_{\sigma(i)}(0) \in 2\pi i\mathbb{Z}.$$

This defines k_i. Shift initial lifts by $m_j \in \mathbb{Z}$. Unique path lifting gives

$$\omega_i'(s) = \omega_i(s) + 2\pi i m_i,$$

and hence

$$\begin{aligned}\omega_i'(1) &= \omega_{\sigma(i)}(0) + 2\pi i(k_i + m_i) \\ &= \omega_{\sigma(i)}'(0) + 2\pi i(k_i + m_i - m_{\sigma(i)}).\end{aligned}$$

This proves (83). If $i_1 \mapsto i_2 \mapsto \cdots \mapsto i_\ell \mapsto i_1$ is a permutation cycle, then

$$\sum_{j=1}^{\ell}(m_{i_j} - m_{\sigma(i_j)}) = 0.$$

The cycle sum of deck shifts is therefore lift-gauge invariant, whereas each summand is not. Relabeling the roots permutes the cycles, so the unordered multiset of cycle sums is canonical.

The total sum has an algebraic expression. If the roots are $z_i(b)$, then

$$\prod_i z_i(b) = (-1)^n c_n(b).$$

The winding of the constant coefficient around zero equals the sum of the root windings and hence $\sum_i k_i$.

B.4 Helical trace on the parameter-position torus

On $S^1_\theta \times S^1_y$, the equation $ny - m\theta \in \pi\mathbb{Z}$ is the inverse image of the two values 0 and π under the group homomorphism

$$(\theta, y) \longmapsto ny - m\theta \pmod{2\pi}.$$

The kernel of the homomorphism with coefficient vector $(-m, n)$ has $d = \gcd(n, m)$ components. Each of the two inverse images is a translate of that kernel, giving $2d$ components in total. A primitive parameterization of any component is

$$s \longmapsto \left(\theta_0 + \frac{n}{d}s,\, y_0 + \frac{m}{d}s\right), \qquad s \in \mathbb{R}/(2\pi\mathbb{Z}),$$

which proves (70). With the ordered torus-link convention declared in Section 8, the union is $T(2n, 2m)$.

B.5 The sign of the simple half-twist

For $w^2 = \tau$ on the positively oriented boundary $\tau = \varepsilon e^{i\theta}$, choose at $\theta = 0$ the ordering $(-\sqrt{\varepsilon}, +\sqrt{\varepsilon})$. As θ increases, both points turn counterclockwise through angle π and exchange. Under the convention that a counterclockwise exchange in the root plane is positive, the braid is σ_1. Reversing the boundary orientation gives σ_1^{-1}. Changing the local coordinate w by an orientation-preserving complex affine map does not change this sign.

This local sign does not determine a global filling. If a parameter disc meets the discriminant in several transverse points, the boundary braid is a product of conjugates of signed half-twists, with order determined by a choice of paths from a basepoint. Holomorphic dependence restricts the possible signs and factorizations; the quasipositive boundary theorem is the relevant classical constraint [17].

B.6 Closure equivalence

Alexander's theorem says that every oriented link in S^3 has a closed-braid representative. Markov's theorem says that two braids have isotopic oriented closures precisely after a sequence generated by braid conjugation and positive or negative stabilization, allowing changes of strand number. Accordingly, a quantity on B_n is not a classical link invariant until the three identities in (100), with its chosen normalization, are proved. If the braid axis is retained, one obtains an annular invariant under a smaller equivalence relation; it must not be relabeled an ordinary S^3 invariant.

C Affine and twisted-quandle calculations

C.1 Conjugation in the affine group

From (84),

$$\begin{aligned}(q,v)^{-1}(p,u)(q,v) &= (q^{-1},-q^{-1}v)(p,u)(q,v)\\ &= \Big(p, q^{-1}(u-v)+q^{-1}pv\Big)\\ &= \Big(p, q^{-1}\big(u+(p-1)v\big)\Big),\end{aligned}$$

which proves (85). The multiplier of the left input is preserved by conjugation. Consequently each fixed-multiplier set is a subquandle, and taking $p = q = t$ gives (86).

The torsion expression can be recognized as the difference between the translation parts of the two affine products. More precisely, define the translation-coordinate projection by $\operatorname{trans}(p,u) = u$. Then

$$\operatorname{trans}\big((p,u)(q,v)\big) - \operatorname{trans}\big((q,v)(p,u)\big) = (1-q)u + (p-1)v.$$

This verifies (87) and its fixed-branch restriction.

C.2 Twisted cocycle and primitive

Write $c = t-1$, $\alpha = t^{-1}$, and let the coefficient action be multiplication by β. For $\phi(x,y) = c(y-x)$, the left-hand side of (89), divided by c, is

$$\beta(y-x) + z - \big(\alpha x + (1-\alpha)y\big) = z - (\alpha+\beta)x + (\alpha+\beta-1)y.$$

The right-hand side is

$$\begin{aligned}&\beta(z-x) + (1-\beta)(z-y)\\ &\quad + \big(\alpha y + (1-\alpha)z\big) - \big(\alpha x + (1-\alpha)z\big)\\ &= z - (\alpha+\beta)x + (\alpha+\beta-1)y.\end{aligned}$$

Thus ϕ is a twisted cocycle for every β.

For $f(x) = Cx$, convention (90) gives

$$\delta_T^+ f(x,y) = C(\alpha - \beta)(y - x).$$

If $\alpha - \beta$ is invertible, choosing $C = c/(\alpha - \beta)$ proves nonresonant exactness. At $\beta = \alpha$, this calculation vanishes for every linear f, but Theorem 11.4 is stronger: it excludes every set-theoretic one-cochain over the finite field.

For completeness, the induction used there follows from $g(\alpha x) = \alpha g(x) + cx$:

$$\begin{aligned} g(\alpha^{j+1}x) &= \alpha g(\alpha^j x) + c\alpha^j x \\ &= \alpha^{j+1} g(x) + (j+1)c\alpha^j x. \end{aligned}$$

C.3 The extension calculation

Relative to the ordered basis of $E = N \oplus X_t$, the operator in (96) is

$$[T_E] = \begin{pmatrix} \alpha & -(t-1) \\ 0 & \alpha \end{pmatrix}, \qquad \det[T_E] = \alpha^2 \neq 0.$$

Thus it defines an Alexander module. With $s(x) = (0, x)$,

$$\begin{gathered} T_E s(x) = (-(t-1)x, \alpha x), \\ (1 - T_E)s(y) = ((t-1)y, (1-\alpha)y). \end{gathered}$$

Their sum is $(\kappa_t(x,y), x \triangleright y)$, which proves (97). Equivalently, the Alexander operation on E is

$$(a, x) * (b, y) = \big(\alpha a + (1-\alpha)b + \kappa_t(x,y), x \triangleright y\big).$$

This is the explicit extension behind Theorem 11.5.

C.4 Two coloring-count checks

The state sum uses crossing weights modified by the Alexander numbering of the source region. The extension theorem makes the total weight zero for every planar coloring, so only the number of colorings remains. Two small cases check the result.

For the positive two-strand braid, use the coloring map

$$B(a,b) = (b, a \triangleright b).$$

Writing $d = b - a$, iteration gives $d_{j+1} = -\alpha d_j$.

The closure of B^2 is the positive Hopf link. Its fixed-point equation forces $(1-\alpha)d = 0$, so $d = 0$ because $t \neq 1$. Hence

$$\Phi_{\kappa_t}(H) = q[0]. \tag{102}$$

The closure of B^3 is the positive trefoil. Its fixed-point equation is $(\alpha^2 - \alpha + 1)d = 0$, giving

$$\Phi_{\kappa_t}(3_1) = \begin{cases} q^2[0], & \alpha^2 - \alpha + 1 = 0, \\ q[0], & \alpha^2 - \alpha + 1 \neq 0. \end{cases} \tag{103}$$

For $q = 3$ and $t = \alpha = 2$, the values are $3[0]$ and $9[0]$. Their difference is entirely the Alexander-quandle coloring count, not a cocycle enhancement.

C.5 Variable-multiplier defect

Let $K = \kappa(x, y)$. From (85),

$$\kappa(x \triangleright y, z) - \kappa(x, z) = \frac{1-r}{q} K,$$
$$\kappa(x \triangleright z, y \triangleright z) = \frac{1}{r} K.$$

Therefore

$$\begin{aligned}\mathfrak{A}_\kappa(x, y, z) &= \left(1 + \frac{1-r}{q} - \frac{1}{r}\right) K \\ &= \frac{(q-r)(r-1)}{qr} K,\end{aligned}$$

which proves (98). The translation coordinate w of z cancels identically; this does not provide the higher coherence absent from the construction.

C.6 The figure-eight word

With $a = (t, 0)$, $b = (1, 1)$, $A = a^{-1}$, and $B = b^{-1}$, multiply the letters of $abbbaBAAB$ from left to right using (84):

Prefix	Multiplier	Translation
a	t	0
ab	t	t
abb	t	$2t$
$abbb$	t	$3t$
$abbba$	t^2	$3t$
$abbbaB$	t^2	$3t - t^2$
$abbbaBA$	t	$3t - t^2$
$abbbaBAA$	1	$3t - t^2$
$abbbaBAAB$	1	$3t - t^2 - 1$

The final translation is $-(t^2 - 3t + 1)$, proving (99) under the declared convention.

References

[1] James W. Alexander. A lemma on systems of knotted curves. *Proceedings of the National Academy of Sciences of the United States of America*, 9(3):93–95, 1923.

[2] Emil Artin. Theorie der zöpfe. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 4:47–72, 1925.

[3] Alan F. Beardon. *The Geometry of Discrete Groups*, volume 91 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1983.

[4] Joan S. Birman. *Braids, Links, and Mapping Class Groups*, volume 82 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ, 1974.

[5] J. Scott Carter, Mohamed Elhamdadi, and Masahico Saito. Twisted quandle homology theory and cocycle knot invariants. *Algebraic & Geometric Topology*, 2:95–135, 2002.

[6] Pierre Dehornoy. Geodesic flow, left-handedness, and templates. *Algebraic & Geometric Topology*, 15(3):1525–1597, 2015.

[7] Pierre Dehornoy and Tali Pinsky. Coding of geodesics and Lorenz-like templates for some geodesic flows. *Ergodic Theory and Dynamical Systems*, 38(3):940–960, 2018.

[8] Charles F. Doran, Terry Gannon, Hossein Movasati, and Khosro M. Shokri. Automorphic forms for triangle groups. *Communications in Number Theory and Physics*, 7(4):689–737, 2013.

[9] Charles Ehresmann. Les connexions infinitésimales dans un espace fibré différentiable. In *Séminaire Bourbaki, 1948/49–1950/51*, volume 1, pages 153–168. Société Mathématique de France, 1952. Exposé no. 24.

[10] Edward Fadell and Lee Neuwirth. Configuration spaces. *Mathematica Scandinavica*, 10:111–118, 1962.

[11] Alexander Grothendieck and Michèle Raynaud. *Revêtements Étales et Groupe Fondamental: Séminaire de Géométrie Algébrique du Bois Marie 1960–61 (SGA 1)*, volume 224 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin, 1971.

[12] Robin Hartshorne. *Algebraic Geometry*, volume 52 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1977.

[13] Morris W. Hirsch. *Differential Topology*, volume 33 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1976.

[14] Cheng Lien Lang and Mong Lung Lang. Arithmetic and geometry of the Hecke groups. *Journal of Algebra*, 460:392–417, 2016.

[15] Jürgen Neukirch. *Algebraic Number Theory*, volume 322 of *Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, Berlin, 1999.

[16] David Rosen. A class of continued fractions associated with certain properly discontinuous groups. *Duke Mathematical Journal*, 21(3):549–563, 1954.

[17] Lee Rudolph. Algebraic functions and closed braids. *Topology*, 22(2):191–202, 1983.

[18] Mingli Yuan. Arithmetic expression geometry i: Foundations—sequential histories, affine flow, torsion, and contact geometry. Mathematical-review manuscript, 2026.

[19] Mingli Yuan. Arithmetic expression geometry ii: Hyperbolic real function theory— horizontal operators, boundary problems, and arithmetic holomorphicity. Mathematical-review manuscript, 2026.

Result	Content	Status
Conformal realization	Every smooth submersion on a surface satisfies (AEG) for an explicit conformal metric.	Proved
Branched pullback	A holomorphic branch of degree m gives a $2m$-pronged essential zero and cone angle $2\pi m$.	Proved
Multi-zero models	Explicit regular AESs have exactly k parallel zero components, or countably many logarithmic lifts.	Proved
$q = 4$ arithmetic network	A Hecke Hauptmodul gives the $\{\infty, 4\}$ zero network, 4π cone nodes, and sign-character descent.	Proved
Sign cover and cusp link	The completed coarse AEG surface of the sign cover is $\mathbb{C}^\times$; retaining the hyperbolic orbifold, its unit tangent bundle is $S^3 \setminus T(2, 4)$ under the standard cusp compactification.	Proved
Hecke geodesic knots	Primitive hyperbolic classes give periodic-orbit knots; the zero dessin is their coding spine, and the cusp linking number is an explicit turn sum using cited classical theorems.	Proved
Relative zero divisor	A supplied monic generically square-free arithmetic polynomial has distinct K-prime, geometric-sheet, Galois, and monodromy levels.	Proved
Register naturality tests	Supplied quadratic and quartic registers give explicit endpoint divisors, trace/pole data, collapse, discriminants, monodromy, and Frobenius cycles.	Proved for supplied registers
General history divisor	A canonical functor sends unrestricted marked histories to arithmetic prime relative divisors without losing required process data.	Open
Incidence theorem	Vertical transversality gives a smooth incidence and a submersion; properness gives a bundle.	Proved with stated hypotheses
Real-tube structure	Compact boundaryless real zero tubes over S^1 with global vertical orientation are unions of tori.	Proved with stated hypotheses
Helical tube	A compact family computes component permutation, logarithmic deck shift, and torus-link boundary traces.	Proved
Singular models	Definite and indefinite Morse events satisfy the AEG equation off an essential singular point.	Proved examples
Root monodromy	Square-free arithmetic-holomorphic polynomial families give proper root coverings and realize every braid.	Proved via Paper II
Affine novelty filter	Stateless scalars collapse to writhe; ordinary affine torsion is exact; resonant finite-field twisted torsion is nonzero but its planar state sum collapses to coloring count.	Proved
Beyond baselines	A Markov-descended, choice-free AEG decoration separates closures not separated by Alexander/Burau baselines.	Open

Table 1: Claim ledger for the main results.